

NOVA Microhypervisor Interface Specification

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The specification could include technical inaccuracies or typographical errors. Additions and changes are periodically made to the information therein; these will be incorporated into new versions of the specification, if any.

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Notation

The key words **must**, **must not**, **required**, **should**, **should not**, **recommended**, **may** and **optional** in this document are to be interpreted as described in RFC 2119 [1].

Throughout this document, the following symbols are used:

- ~ Indicates that the value of this parameter or field is **unknown**. The microhypervisor cannot ensure that the value does not leak information across protection domain boundaries.
- ⚡ Indicates that the value of this parameter or field is **undefined**. The microhypervisor ensures that the value does not leak information across protection domain boundaries. Future versions of this specification may define a value for the parameter or field.
- Indicates that the value of this parameter or field is **ignored**. Future versions of this specification may define a meaning for the parameter or field.
- ≡ Indicates that the value of this parameter or field is **unchanged**. The microhypervisor preserves the value across hypercalls.

Part I

Introduction

1 System Architecture

The NOVA OS Virtualization Architecture [2] (NOVA) facilitates the coexistence of multiple legacy guest operating systems and a user-mode host framework on a single platform. The core system leverages hardware virtualization technology provided by modern x86 or Arm platforms and comprises the NOVA microhypervisor and one or more Virtual-Machine Monitors (VMMs).

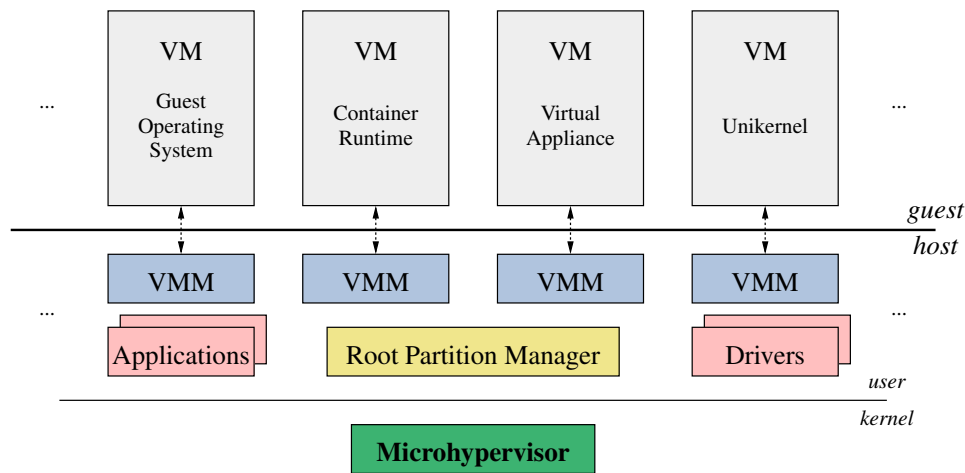


Figure 1.1: System Architecture

Figure 1.1 shows the structure of the system. The microhypervisor is the only component executing in privileged host/kernel mode. It isolates the various user-mode components, including the virtual-machine monitors, from one another by placing them in different protection domains in unprivileged host/user mode. Each legacy guest operating system runs in its own virtual-machine environment in guest mode and is therefore isolated from the other components.

Besides spatial and temporal isolation, the microhypervisor also provides mechanisms for partitioning and delegation of platform resources, such as CPU time, physical memory, I/O ports and hardware interrupts and for establishing communication channels and signaling between different protection domains.

The virtual-machine monitors handle virtualization events and implement virtual devices that enable legacy guest operating systems to function in the same manner as they would on bare-metal hardware. Providing this functionality outside the microhypervisor in the VMMs reduces the size of the trusted computing base significantly for all components that do not require virtualization support.

The architecture and interfaces of the VMM and the user-mode host framework are not described in this document.

Part II

Basic Abstractions

2 Kernel Objects

2.1 Protection Domain

1. The **Protection Domain (PD)** is a unit of protection and spatial isolation.
2. Access to a **Protection Domain** is controlled by a **PD Capability** ($CAP_{OBJ_{PD}}$).
3. Different types of spaces can be created for a **PD** – see Section 4.2 for details regarding availability and size.
4. Spaces store **Capabilities (CAPs)** for kernel objects or platform resources that **Execution Contexts** in that **Protection Domain** can access.

The following subsections describe each space in more detail.

2.1.1 Object Space

1. Only one **Object Space** (SPC_{OBJ}) can be created per **Protection Domain**.
2. Access to an **Object Space** is controlled by an **Object Space Capability** ($CAP_{OBJ_{OBJ}}$).
3. Each hypercall invoked by a **Host Execution Context** explicitly specifies an **Object Capability Selector** to designate the kernel object on which it operates.
4. The **Object Capability Selector** (SEL_{OBJ}) serves as index into the **EC's Object Space** and selects a slot that contains either a **Null Capability** (CAP_0) or an **Object Capability** (CAP_{OBJ}) that refers to a kernel object with associated access permissions.

2.1.2 Host Space

1. Only one **Host Space** (SPC_{HST}) can be created per **Protection Domain**.
2. Access to a **Host Space** is controlled by a **Host Space Capability** ($CAP_{OBJ_{HST}}$).
3. Each memory operation issued by a **Host Execution Context** implicitly uses the page number of the accessed **Host-Virtual Address (HVA)** as **Host Capability Selector**: $SEL_{HST} = HVA \gg 12$.
4. The **Host Capability Selector** (SEL_{HST}) serves as index into the **EC's Host Space** and selects a slot that contains either a **Null Capability** (CAP_0) or a **Memory Capability** (CAP_{MEM}) that refers to a 4 KiB page frame in physical memory with associated access permissions.

2.1.3 Guest Space

1. Multiple **Guest Spaces** (SPC_{GST}) can be created per **Protection Domain**.
2. Access to a **Guest Space** is controlled by a **Guest Space Capability** ($CAP_{OBJ_{GST}}$).
3. Each memory operation issued by a **Guest Execution Context** implicitly uses the page number of the accessed **Guest-Physical Address (GPA)** as **Guest Capability Selector**: $SEL_{GST} = GPA \gg 12$.
4. The **Guest Capability Selector** (SEL_{GST}) serves as index into the **EC's Guest Space** and selects a slot that contains either a **Null Capability** (CAP_0) or a **Memory Capability** (CAP_{MEM}) that refers to a 4 KiB page frame in physical memory with associated access permissions.

2.1.4 DMA Space

1. Multiple **DMA Spaces** (SPC_{DMA}) can be created per **Protection Domain**.
2. Access to a **DMA Space** is controlled by a **DMA Space Capability** ($CAP_{OBJ_{DMA}}$).
3. Each **DMA** operation issued by a device implicitly uses the page number of the accessed **DMA-Virtual Address** (DVA) as **DMA Capability Selector**: $SEL_{DMA} = DVA \gg 12$.
4. The **DMA Capability Selector** (SEL_{DMA}) serves as index into the device's **DMA Space** and selects a slot that contains either a **Null Capability** (CAP_0) or a **Memory Capability** (CAP_{MEM}) that refers to a 4 KiB page frame in physical memory with associated access permissions.

2.1.5 PIO Space

1. Multiple **PIO Spaces** (SPC_{PIO}) can be created per **Protection Domain**.
2. Access to a **PIO Space** is controlled by a **PIO Space Capability** ($CAP_{OBJ_{PIO}}$).
3. Each **PIO** access (IN/OUT instruction) issued by an **Execution Context** (EC) implicitly uses the number of the accessed I/O port as **PIO Capability Selector**.
4. The **PIO Capability Selector** (SEL_{PIO}) serves as index into the EC's **PIO Space** and selects a slot that contains either a **Null Capability** (CAP_0) or a **PIO Capability** (CAP_{PIO}) that refers to the I/O port SEL_{PIO} with associated access permissions.

2.1.6 MSR Space

1. Multiple **MSR Spaces** (SPC_{MSR}) can be created per **Protection Domain**.
2. Access to an **MSR Space** is controlled by an **MSR Space Capability** ($CAP_{OBJ_{MSR}}$).
3. Each **MSR** access (RDMSR/WRMSR instruction) issued by an **Execution Context** (EC) implicitly uses the number of the accessed **MSR** as **MSR Capability Selector**.
4. The **MSR Capability Selector** (SEL_{MSR}) serves as index into the EC's **MSR Space** and selects a slot that contains either a **Null Capability** (CAP_0) or an **MSR Capability** (CAP_{MSR}) that refers to the **MSR** SEL_{MSR} with associated access permissions.

2.2 Execution Context

1. The **Execution Context (EC)** is an abstraction for an activity within a **PD**.
2. Access to an **Execution Context** is controlled by an **EC Capability** ($CAP_{OBJ_{EC}}$).
3. An **EC** is permanently bound to one **CPU**.
4. An **EC** contains architecture-dependent state, such as
 - Central Processing Unit (**CPU**) registers
 - Floating Point Unit (**FPU**) registers (optionally)

The following subsections provide more details for each type of **Execution Context**.

2.2.1 Host Execution Context

1. There exist two types of **Host Execution Context** (EC_{HST}):
 - Local Threads – these may have **PTs** (but no **SCs**) bound to it.
 - Global Threads – these may have an **SC** (but no **PTs**) bound to it.
2. A **Host Execution Context** has a **UTCB** that enables it to perform regular **IPC**.
3. Upon creation, a **Host Execution Context** is permanently bound to the following **required** spaces of its **PD**:
 - The first **Object Space***
 - The first **Host Space***
 - The first **PIO Space**[†]
4. A **Host Execution Context** cannot be reassigned to any spaces.

2.2.2 Guest Execution Context

1. There exists one type of **Guest Execution Context** (EC_{GST}):
 - Virtual CPUs – these may have an **SC** (but no **PTs**) bound to it.
2. A **Guest Execution Context** does not have a **UTCB**.
3. Upon creation, a **Guest Execution Context** is permanently bound to the following **required** spaces of its **PD**:
 - The first **Object Space***
 - The first **Host Space***
4. Upon any architectural/microhypervisor event, a **Guest Execution Context** may be (re)assigned to the following spaces of its **PD**, by setting the **SPACES** bit in the **MTD** and providing selectors that refer to capabilities with **ASSIGN** permission for these spaces in the respective **UTCB** fields during **ipc_reply**:
 - Any **Guest Space**[†]
 - Any **PIO Space**[†]
 - Any **MSR Space**[†]

The handler of the **STARTUP** event must perform the initial assignment of these spaces.

*Because only one such space can be created for a **PD**, it is also the only such space.

[†]Only on architectures, where this type of space is **available**.

2.3 Scheduling Context

1. The **Scheduling Context (SC)** is a unit of prioritization and temporal isolation.
2. Access to a **Scheduling Context** is controlled by an **SC Capability** ($CAP_{OBJ_{SC}}$).
3. An **SC** is permanently bound to one **CPU**.
4. An **SC** is permanently bound to the **EC** for which it was created.
5. Donation allows another **EC** to consume the budget of the **SC** for the duration of the donation.
6. A scheduling context comprises the following state:
 - Reference to bound **EC** (2.2)
 - Class Of Service (**COS**)
 - Priority – numerically higher priorities always preempt numerically lower priorities
 - Budget – time after which the **SC** can be preempted by an **SC** with the same priority

2.4 Portal

1. A **Portal (PT)** represents a dedicated entry point into the **PD** for which the portal was created.
2. Access to a **Portal** is controlled by a **PT Capability** ($CAP_{OBJ_{PT}}$).
3. A **PT** is permanently bound to the **EC** for which it was created.
4. A portal comprises the following state:
 - Reference to bound **EC** (2.2)
 - **Message Transfer Descriptor (MTD)** (4.4)
 - Entry Instruction Pointer (**IP**)
 - Portal Identifier (**PID**)

2.5 Semaphore

1. A **Semaphore (SM)** provides a means to synchronize execution and interrupt delivery by selectively blocking and unblocking **Execution Contexts (ECs)**.
2. Access to a **Semaphore** is controlled by a **SM Capability** ($CAP_{OBJ_{SM}}$).

2.6 Device Context

1. A **Device Context (DC)** provides management information for a hardware device.
2. Access to a **Device Context** is controlled by a **DC Capability** ($CAP_{OBJ_{DC}}$).
3. A **Device Context** comprises the following state:
 - Topology of the device, such as **PCI Segment/Bus/Device/Function (SBDF)**
 - Reference to the component that performs **DMA** translation for the device, such as **IOMMU** or **SMMU**
 - Reference to the component that performs **MSI** translation for the device, such as **IOMMU** or **GITS**
 - Auxiliary information

3 Hardware Resources

3.1 System Time Counter

The system time is represented by an unsigned 64-bit [System Time Counter \(STC\)](#) with the following properties:

1. The [STC](#) starts with a power-on value of 0.
2. Subsequent reads of the [STC](#) return a higher value that reflects the platform uptime.
3. While the platform is in a shallow sleep state, the [STC](#) retains its current value.
4. While the platform is running, the [STC](#) monotonically increments at a fixed frequency, which is conveyed in the [Hypervisor Information Page \(HIP\)](#).
5. The [STC](#) and its frequency are synchronized across all [CPUs](#). Applications can use both values to convert between system time and wall clock time.
6. Applications can obtain the current [STC](#) value as follows:

Arm: By reading CNTVCT_EL0 via the MRS instruction [3].

x86: By reading IA32_TSC via the RDTSC instruction [4, 5].

Part III

Application Programming Interface

4 Data Types

4.1 Capability

Capabilities are communicable, unforgeable tokens of authority. They consist of a reference to a resource coupled with access permissions. **Capabilities** are opaque and immutable for applications – they cannot be inspected or modified directly; instead applications refer to a **Capability** (**CAP**) via a **Capability Selector** (**SEL**).

4.1.1 Null Capability

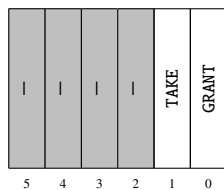
A **Null Capability** (**CAP₀**) does not refer to anything and carries no permissions.

4.1.2 Object Capability

An **Object Capability** (**CAP_{OBJ}**) is stored in the **Object Space** of a **PD** and refers to a kernel object.

4.1.2.1 Object Space Capability

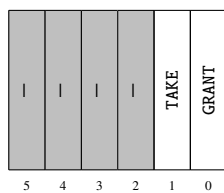
An **Object Space Capability** (**CAP_{OBJ_{OBJ}}**) refers to an **Object Space** and carries the following permissions:



GRANT If set, **ctrl_pd** can use that **Object Space** as destination.
TAKE If set, **ctrl_pd** can use that **Object Space** as source.

4.1.2.2 Host Space Capability

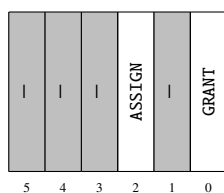
A **Host Space Capability** (**CAP_{OBJ_{HST}}**) refers to a **Host Space** and carries the following permissions:



GRANT If set, **ctrl_pd** can use that **Host Space** as destination.
TAKE If set, **ctrl_pd** can use that **Host Space** as source.

4.1.2.3 Guest Space Capability

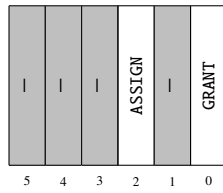
A **Guest Space Capability** (**CAP_{OBJ_{GST}}**) refers to a **Guest Space** and carries the following permissions:



GRANT If set, **ctrl_pd** can use that **Guest Space** as destination.
ASSIGN If set, **ipc_reply** can assign an **EC_{GST}** to that **Guest Space**.

4.1.2.4 DMA Space Capability

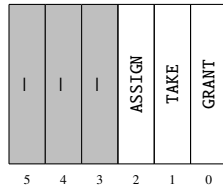
A **DMA Space Capability** ($CAP_{OBJ_{DMA}}$) refers to a **DMA Space** and carries the following permissions:



GRANT	If set, ctrl_pd can use that DMA Space as destination.
ASSIGN	If set, assign_dev can assign a device to that DMA Space .

4.1.2.5 PIO Space Capability

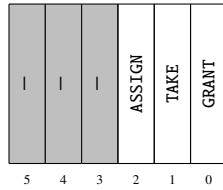
A **PIO Space Capability** ($CAP_{OBJ_{PIO}}$) refers to a **PIO Space** and carries the following permissions:



GRANT	If set, ctrl_pd can use that PIO Space as destination.
TAKE	If set, ctrl_pd can use that PIO Space as source.
ASSIGN	If set, ipc_reply can assign an EC_{GST} to that PIO Space .

4.1.2.6 MSR Space Capability

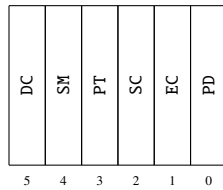
An **MSR Space Capability** ($CAP_{OBJ_{MSR}}$) refers to an **MSR Space** and carries the following permissions:



GRANT	If set, ctrl_pd can use that MSR Space as destination.
TAKE	If set, ctrl_pd can use that MSR Space as source.
ASSIGN	If set, ipc_reply can assign an EC_{GST} to that MSR Space .

4.1.2.7 Protection Domain Capability

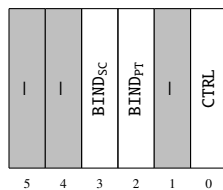
A **PD Capability** ($CAP_{OBJ_{PD}}$) refers to a **Protection Domain (PD)** and carries the following permissions:



PD	If set, create_pd permitted.
EC	If set, create_ec permitted.
SC	If set, create_sc permitted.
PT	If set, create_pt permitted.
SM	If set, create_sm permitted.
DC	If set, create_dc permitted.

4.1.2.8 Execution Context Capability

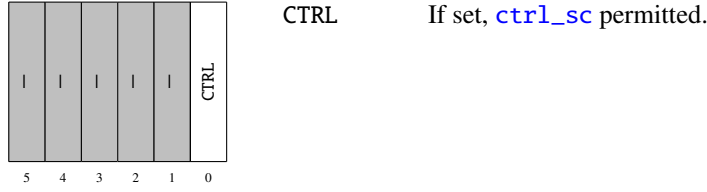
An **EC Capability** ($CAP_{OBJ_{EC}}$) refers to an **Execution Context (EC)** and carries the following permissions:



CTRL	If set, ctrl_ec permitted.
BIND _{PT}	If set, create_pt can bind a Portal (PT) to the EC .
BIND _{SC}	If set, create_sc can bind a Scheduling Context (SC) to the EC .

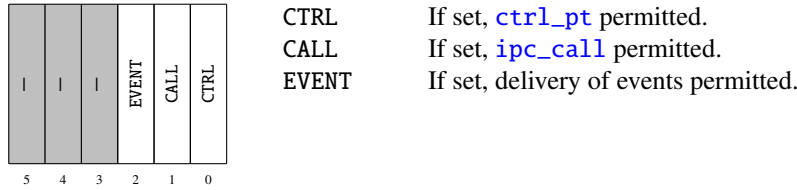
4.1.2.9 Scheduling Context Capability

An **SC Capability** ($CAP_{OBJ_{SC}}$) refers to a **Scheduling Context (SC)** and carries the following permissions:



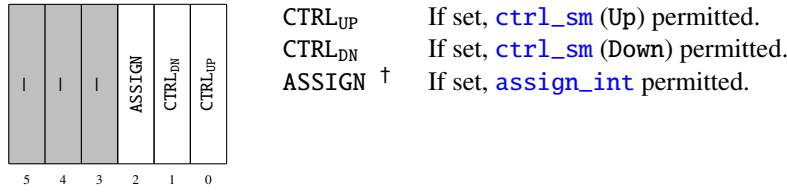
4.1.2.10 Portal Capability

A **PT Capability** ($CAP_{OBJ_{PT}}$) refers to a **Portal (PT)** and carries the following permissions:



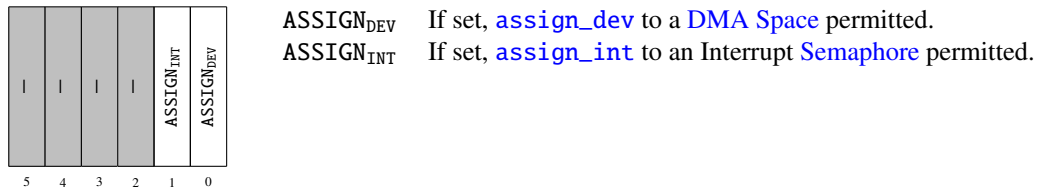
4.1.2.11 Semaphore Capability

An **SM Capability** ($CAP_{OBJ_{SM}}$) refers to a **Semaphore (SM)** and carries the following permissions:



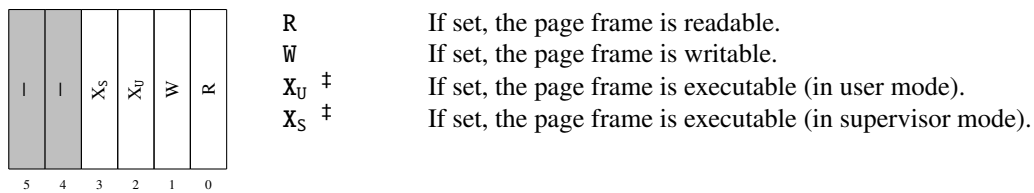
4.1.2.12 Device Context Capability

A **DC Capability** ($CAP_{OBJ_{DC}}$) refers to a **Device Context (DC)** and carries the following permissions:



4.1.3 Memory Capability

A **Memory Capability** (CAP_{MEM}) is stored in a **Host Space**, **Guest Space** or **DMA Space** of a **PD**, refers to a 4 KiB page frame, and carries the following permissions:

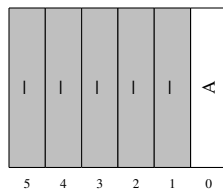


[†]This permission bit is only defined for interrupt semaphores.

[‡]If the hardware supports only combined execute permissions (X) for both modes, then $X = X_U \vee X_S$.

4.1.4 PIO Capability

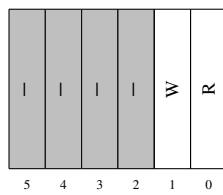
A **PIO Capability** (CAP_{PIO}) is stored in a **PIO Space** of a **PD**, refers to an I/O port, and carries the following permissions:



A If set, the I/O port is accessible (via IN/OUT).

4.1.5 MSR Capability

An **MSR Capability** (CAP_{MSR}) is stored in an **MSR Space** of a **PD**, refers to a Model-Specific Register (**MSR**), and carries the following permissions:



R If set, the **MSR** is readable (via RDMSR).
W If set, the **MSR** is writable (via WRMSR).

4.2 Capability Selector

A [Capability Selector \(SEL\)](#) is an application-visible unsigned number that serves as an index into a space of an [Execution Context](#) to select a [Capability \(CAP\)](#). The following subsections provide more details.

4.2.1 Object Capability Selector

1. [Object Spaces \(SPC_{OBJ}\)](#) are unavailable if $SBW_{OBJ} = 0$. No systems currently report this.
2. Otherwise, a valid [Object Capability Selector \(SEL_{OBJ}\)](#) is in range $[0, 2^{SBW_{OBJ}})$.

4.2.2 Host Capability Selector

1. [Host Spaces \(SPC_{HST}\)](#) are unavailable if $SBW_{HST} = 0$. No systems currently report this.
2. Otherwise, a valid [Host Capability Selector \(SEL_{HST}\)](#) is in range $[0, 2^{SBW_{HST}})$. $\implies$ valid [HVA](#) range

4.2.3 Guest Capability Selector

1. [Guest Spaces \(SPC_{GST}\)](#) are unavailable if $SBW_{GST} = 0$. Systems without [CPU](#) virtualization may report this.
2. Otherwise, a valid [Guest Capability Selector \(SEL_{GST}\)](#) is in range $[0, 2^{SBW_{GST}})$. $\implies$ valid [GPA](#) range

4.2.4 DMA Capability Selector

1. [DMA Spaces \(SPC_{DMA}\)](#) are unavailable if $SBW_{DMA} = 0$. Systems without I/O virtualization may report this.
2. Otherwise, a valid [DMA Capability Selector \(SEL_{DMA}\)](#) is in range $[0, 2^{SBW_{DMA}})$. $\implies$ valid [DVA](#) range

4.2.5 PIO Capability Selector

1. [PIO Spaces \(SPC_{PIO}\)](#) are unavailable if $SBW_{PIO} = 0$. Non-x86 systems may report this.
2. Otherwise, a valid [PIO Capability Selector \(SEL_{PIO}\)](#) is in range $[0, 2^{SBW_{PIO}})$. $\implies$ valid [PIO](#) range

4.2.6 MSR Capability Selector

1. [MSR Spaces \(SPC_{MSR}\)](#) are unavailable if $SBW_{MSR} = 0$. Non-x86 systems may report this.
2. Otherwise, a valid [MSR Capability Selector \(SEL_{MSR}\)](#) is in range $[0, 2^{SBW_{MSR}})$. $\implies$ valid [MSR](#) range

4.3 User Thread Control Block

A **User Thread Control Block (UTCb)** has a size of one memory page (4 KiB) and is mapped in the **Host Space** of its associated **Host Execution Context**. Because a **UTCb** is allocated and owned by the microhypervisor, it cannot be delegated using `ctrl_pd`.

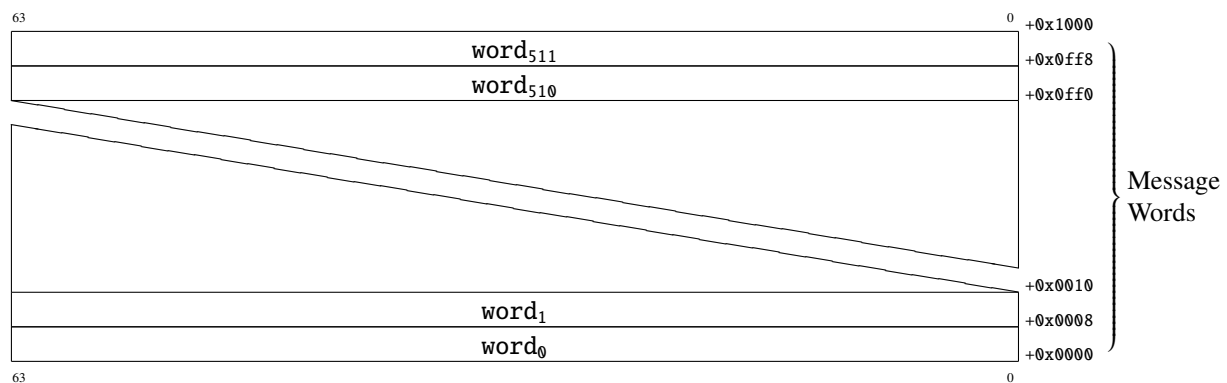
To ensure proper visibility of loads and stores with relaxed memory ordering, application programs are expected to access a **UTCb** only from the **Host Execution Context** to which that **UTCb** is bound.

4.3.1 Regular Layout

During regular **IPC** (see 4.4.1), the **UTCb** is used for data transfer.

The data transfer from one **UTCb** to another **UTCb** is defined as follows:

- The data transfer is performed by the **CPU** on which the caller **EC** and callee **EC** execute.
- The data transfer uses the regular layout with 512 message words (see below).
- The data is copied from low words to high words, beginning with `word0`.
- The granularity of the loads and stores used for copying is **undefined**.
- Loads from and stores to the **UTCb** are **non-atomic** and use **relaxed** memory ordering.



4.3.2 Architectural Layout

During architectural **IPC** (see 4.4.2), the **UTCb** is used for state transfer.

The state transfer between the architectural registers and a **UTCb** is defined as follows:

- The state transfer is performed by the **CPU** on which the affected **EC** and callee **EC** execute.
- The state transfer uses the architectural layout (**Arm**, **x86**).
- The state is copied between architectural registers and the **UTCb** in an **undefined** order.
- The granularity of the loads and stores used for copying is **undefined**.
- Loads from and stores to the **UTCb** are **non-atomic** and use **relaxed** memory ordering.

4.4 Message Transfer Descriptor

4.4.1 Regular IPC

For regular [Inter-Process Communication \(IPC\)](#), the [Message Transfer Descriptor \(MTD\)](#) is provided by the sender, passed to the receiver, and uses the following layout:



The [MTD](#) controls the data transfer (see 4.3.1) as shown in Figure 4.1:

- During [ipc_call](#), it specifies the number of message words to transfer from the [UTCB](#) of the caller [EC](#) (sender) to the [UTCB](#) of the callee [EC](#) (receiver).
- During [ipc_reply](#), it specifies the number of message words to transfer from the [UTCB](#) of the callee [EC](#) (sender) to the [UTCB](#) of the caller [EC](#) (receiver).

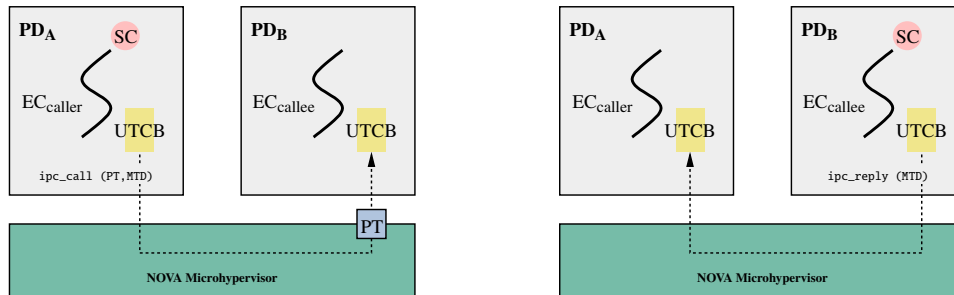


Figure 4.1: Regular [IPC](#)

4.4.2 Architectural IPC

For exceptions and intercepts, the [Message Transfer Descriptor \(MTD\)](#) is provided by the architectural event-specific portal ([Arm, x86](#)) or sender, passed to the receiver, and uses an architectural bitfield layout ([Arm, x86](#)):

- If a bit is 0, then the microhypervisor does **not** transmit the architectural state associated with that bit.
- If a bit is 1, then the microhypervisor transmits the architectural state associated with that bit.

The [MTD](#) controls the state transfer (see 4.3.2) as shown in Figure 4.2:

- During an exception/intercept, it specifies the subset of registers to transfer from the architectural state of the affected [EC](#) (sender) to the [UTCB](#) of the callee [EC](#) (receiver).
- During [ipc_reply](#), it specifies the subset of registers to transfer from the [UTCB](#) of the callee [EC](#) (sender) to the architectural state of the affected [EC](#) (receiver).

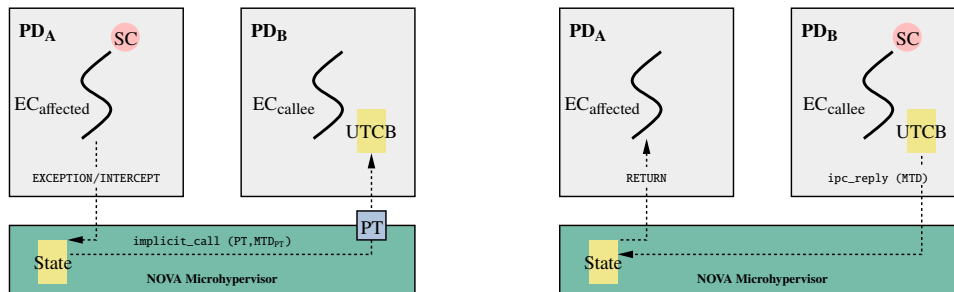
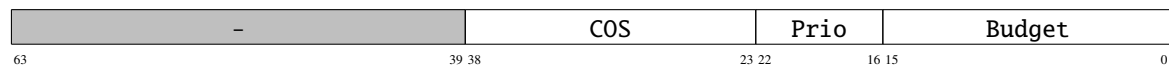


Figure 4.2: Architectural [IPC](#)

4.5 Scheduling Context Descriptor

The [Scheduling Context Descriptor \(SCD\)](#) describes the configuration of a [Scheduling Context \(SC\)](#).



The fields are defined as follows:

Budget

Specifies the scheduling budget in milliseconds – must be > 0.

Prio

Specifies the scheduling priority – must be > 0.

COS

Specifies the Class Of Service – valid values depend on architectural [COS](#) support ([Arm](#), [x86](#)):

- If [COS](#) is not supported, then this field must be 0.
- If [COS](#) is supported, then this field must be < [COS](#)_{NUM}.

5 Hypercalls

5.1 Definitions

5.1.1 Hypercall Numbers

Each hypercall is identified by a unique number. The following hypercalls are currently defined:

Number	Hypercall	Section
0x0	ipc_call	5.2.1
0x1	ipc_reply	5.2.2
0x2	create_pd	5.3.1
0x3	create_ec	5.3.2
0x4	create_sc	5.3.3
0x5	create_pt	5.3.4
0x6	create_sm	5.3.5
0x7	create_dc	5.3.6
0x8	ctrl_pd	5.4.1
0x9	ctrl_ec	5.4.2
0xa	ctrl_sc	5.4.3
0xb	ctrl_pt	5.4.4
0xc	ctrl_sm	5.4.5
0xd	ctrl_hw	5.5.1
0xe	assign_dev	5.5.2
0xf	assign_int	5.5.3

5.1.2 Status Codes

Hypercalls return a status code to indicate success or failure. The following status codes are currently defined:

Number	Status Code	Description
0x0	SUCCESS	Operation Successful
0x1	TIMEOUT	Operation Timeout
0x2	ABORTED	Operation Abort
0x3	OVRFLOW	Operation Overflow
0x4	BAD_HYP	Invalid Hypercall
0x5	BAD_CAP	Invalid Capability
0x6	BAD_PAR	Invalid Parameter
0x7	BAD_FTR	Invalid Feature
0x8	BAD_CPU	Invalid CPU Number
0x9	BAD_DEV	Invalid Device ID
0xa	MEM_OBJ	Insufficient Memory (Object Creation)
0xb	MEM_CAP	Insufficient Memory (Capability Creation)
≥0xc	<i>reserved for future use</i>	

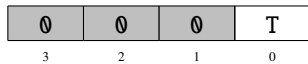
5.2 Communication

5.2.1 IPC Call

Parameters:

```
status = ipc_call (SELOBJ pt,          // Portal
                  MTD& mtd)           // Message Transfer Descriptor
```

Flags:



Description:

Sends a message from **EC_{CURRENT}** (caller) to the **EC** (callee) to which the specified **Portal (PT)** is bound.

Prior to the hypercall:

- **SPC_{OBJ_{CURRENT}}[pt]** must refer to a **PT Capability (CAP_{OBJ_{PT}})** with permission **CALL**.

If the hypercall completed successfully:

- If **T=0 (No Timeout)**: If the callee **EC** was still busy handling a prior **ipc_call**, then the caller **EC** has helped run that prior **ipc_call** to completion, i.e. until the callee **EC** became available again.
- The microhypervisor has transferred a message from the **UTCB** of the caller **EC** to the **UTCB** of the callee **EC**. The content of that message is defined by the **MTD mtd**, which has been passed from the caller **EC** to the callee **EC**.
- The hypercall returns once the callee **EC** has invoked an **ipc_reply**. Upon return, the **UTCB** of the caller **EC** and the **mtd** parameter have been updated by the reply message.
- The Current Scheduling Context (**SC_{CURRENT}**) has been donated to the callee **EC** upon **ipc_call** and returned back upon **ipc_reply**, thereby accounting the entire handling of the request to **SC_{CURRENT}**.

Status:

SUCCESS

- The hypercall completed successfully.

BAD_CAP

- **SPC_{OBJ_{CURRENT}}[pt]** did not refer to a **PT Capability (CAP_{OBJ_{PT}})** or that capability had insufficient permissions.

BAD_CPU

- Caller **EC** and callee **EC** are on different **CPU**s.

TIMEOUT

- If **T=1 (Timeout)**: The callee **EC** is still busy handling a prior **ipc_call**.

ABORTED

- The callee **EC** is dead and the operation aborted.

5.2.2 IPC Reply

Parameters:

```
pid = ipc_reply (MTD & mtd)           // Message Transfer Descriptor
```

Flags:

0	0	0	0
3	2	1	0

Description:

Sends a reply message from **EC_{CURRENT}** (callee) back to the caller **EC** (if one exists) and subsequently waits for the next incoming message.

If the hypercall completed successfully:

- If a caller **EC** exists:
 - The microhypervisor has transferred a reply message from the **UTCB** of the callee **EC** back to the **UTCB** of the caller **EC**.
 - The content of that reply message is defined by the **MTD** **mtd**, which has been passed from the callee **EC** back to the caller **EC**.
 - The Current Scheduling Context (**SC_{CURRENT}**) that had been donated to the callee **EC** upon **ipc_call** has been returned back to the caller **EC**.
- **EC_{CURRENT}** blocks until the next incoming message arrives on any **Portal (PT)** bound to it.

Status:

This hypercall does not return directly.

Instead, when the next message arrives via a subsequent **ipc_call** to any **Portal (PT)** bound to the callee **EC**:

- The microhypervisor passes the Portal Identifier (**PID**) of the called **PT** to the callee **EC**.
- The **UTCB** of the callee **EC** and the **mtd** parameter have been updated by the incoming message.
- Execution of the callee **EC** continues at the Instruction Pointer (**IP**) configured in the called **PT**.

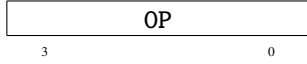
5.3 Object Creation

5.3.1 Create Protection Domain

Parameters:

```
status = create_pd (SEL_OBJ sel,          // Created PD or Space
                   SEL_OBJ pd)           // Owner PD
```

Flags:



Description:

Creates a new **Protection Domain (PD)** or an empty new space for a **Protection Domain**.

Prior to the hypercall:

- $SPC_{OBJ_CURRENT}[sel]$ must refer to a **Null Capability (CAP_0)**.
- $SPC_{OBJ_CURRENT}[pd]$ must refer to a **PD Capability (CAP_{OBJ_PD})** with permission PD.

If the hypercall completed successfully:

- If **OP=0 (Protection Domain)**:
 - A new **Protection Domain (PD)** has been created.
 - $SPC_{OBJ_CURRENT}[sel]$ refers to a CAP_{OBJ_PD} for the created **Protection Domain** with the **permissions** inherited from $SPC_{OBJ_CURRENT}[pd]$.
- If **OP=1 (Object Space)**:
 - A new **Object Space (SPC_{OBJ})** has been created for the **PD** referred to by $SPC_{OBJ_CURRENT}[pd]$.
 - $SPC_{OBJ_CURRENT}[sel]$ refers to a CAP_{OBJ_OBJ} for the created **Object Space** with all **defined** permissions.
- If **OP=2 (Host Space)**:
 - A new **Host Space (SPC_{HST})** has been created for the **PD** referred to by $SPC_{OBJ_CURRENT}[pd]$.
 - $SPC_{OBJ_CURRENT}[sel]$ refers to a CAP_{OBJ_HST} for the created **Host Space** with all **defined** permissions.
- If **OP=3 (Guest Space)**:
 - A new **Guest Space (SPC_{GST})** has been created for the **PD** referred to by $SPC_{OBJ_CURRENT}[pd]$.
 - $SPC_{OBJ_CURRENT}[sel]$ refers to a CAP_{OBJ_GST} for the created **Guest Space** with all **defined** permissions.
- If **OP=4 (DMA Space)**:
 - A new **DMA Space (SPC_{DMA})** has been created for the **PD** referred to by $SPC_{OBJ_CURRENT}[pd]$.
 - $SPC_{OBJ_CURRENT}[sel]$ refers to a CAP_{OBJ_DMA} for the created **DMA Space** with all **defined** permissions.
- If **OP=5 (PIO Space)**:
 - A new **PIO Space (SPC_{PIO})** has been created for the **PD** referred to by $SPC_{OBJ_CURRENT}[pd]$.
 - $SPC_{OBJ_CURRENT}[sel]$ refers to a CAP_{OBJ_PIO} for the created **PIO Space** with all **defined** permissions.
- If **OP=6 (MSR Space)**:
 - A new **MSR Space (SPC_{MSR})** has been created for the **PD** referred to by $SPC_{OBJ_CURRENT}[pd]$.
 - $SPC_{OBJ_CURRENT}[sel]$ refers to a CAP_{OBJ_MSR} for the created **MSR Space** with all **defined** permissions.
- The resources for the created object were accounted to the **PD** referred to by $SPC_{OBJ_CURRENT}[pd]$.

Status:

SUCCESS

- The hypercall completed successfully.

ABORTED

- The **PD** designated by $SPC_{OBJ_{CURRENT}}[pd]$ is being destructed concurrently.
- If $OP > 0$: The space could not bind to the **PD** designated by $SPC_{OBJ_{CURRENT}}[pd]$.

BAD_CAP

- $SPC_{OBJ_{CURRENT}}[sel]$ did not refer to a **Null Capability** (CAP_0).
- $SPC_{OBJ_{CURRENT}}[pd]$ did not refer to a **PD Capability** ($CAP_{OBJ_{PD}}$) or that capability had insufficient permissions.

BAD_PAR

- The requested operation (OP) is invalid.

BAD_FTR

- The requested space type is not supported by the hardware architecture.

MEM_OBJ

- The **Protection Domain** referred to by $SPC_{OBJ_{CURRENT}}[pd]$ had insufficient memory resources for object creation.

MEM_CAP

- The **Protection Domain** to which $SPC_{OBJ_{CURRENT}}$ belongs had insufficient memory resources for capability creation.

5.3.2 Create Execution Context

Parameters:

```
status = create_ec (SEL_OBJ sel,          // Created EC
                   SEL_OBJ pd,          // Owner PD
                   SEL_HST hvp,        // Host-Virtual Page Number
                   SEL_EVT evt,        // Event Selector Base
                   UINT_16 cpu,        // CPU Number
                   UINT_PTR sp)        // Initial Stack Pointer
```

Flags:

0	F	T	G
3	2	1	0

Description:

Creates a new [Execution Context \(EC\)](#).

Prior to the hypercall:

- $SPC_{OBJ_{CURRENT}}[sel]$ must refer to a [Null Capability \(CAP₀\)](#).
- $SPC_{OBJ_{CURRENT}}[pd]$ must refer to a [PD Capability \(CAP_{OBJ_{PD}}\)](#) with permission EC.
- If **G=0**: All spaces [required](#) for a [Host Execution Context](#) must have been created for the owner [PD](#).
- If **G=1**: All spaces [required](#) for a [Guest Execution Context](#) must have been created for the owner [PD](#).

If the hypercall completed successfully:

- If **G=0 (Host Execution Context)**:
 - A new [Host Execution Context \(EC_{HST}\)](#) has been created.
 - The microhypervisor has allocated a [UTCB](#) for [EC_{HST}](#) and mapped it at [HVA hvp](#) << 12.
 - If **T=0 (Local Thread)**: [Portals \(PTs\)](#) may subsequently be bound to [EC_{HST}](#), which will run whenever any of those bound portals is called.
 - If **T=1 (Global Thread)**: [EC_{HST}](#) will generate a [STARTUP](#) event the first time a [Scheduling Context \(SC\)](#) is bound to it.
- If **G=1 (Guest Execution Context)**:
 - A new [Guest Execution Context \(EC_{GST}\)](#) has been created.
 - The microhypervisor has allocated a [vAPIC](#) page for [EC_{GST}](#) and mapped it at [HVA hvp](#) << 12. On non-Intel architectures, the parameter [hvp](#) was ignored.
 - [EC_{GST}](#) will generate a [STARTUP](#) event the first time a [Scheduling Context \(SC\)](#) is bound to it.
 - If **T=0 (Virtual CPU)**: The virtual CPU uses no time adjustment.
 - If **T=1 (Virtual CPU)**: The virtual CPU uses time offsetting.
- The created [EC](#) will be able to use [FPU](#) instructions only if **F=1 (FPU)**. Otherwise any [FPU](#) access by that [EC](#) will generate an architecture-specific exception.
- The created [EC](#) has its [Stack Pointer \(SP\)](#) set to [sp](#) and its [Event Selector Base \(SEL_{EVNT}\)](#) set to [evt](#).[†]
- The created [EC](#) is permanently bound to the [CPU](#) [cpu](#) and to the required spaces of the [PD](#) referred to by $SPC_{OBJ_{CURRENT}}[pd]$.
- The resources for the created [EC](#) were accounted to the [PD](#) referred to by $SPC_{OBJ_{CURRENT}}[pd]$.
- $SPC_{OBJ_{CURRENT}}[sel]$ refers to a [CAP_{OBJ_{EC}}](#) for the created [EC](#) with all [defined](#) permissions.

[†]The microhypervisor sets these values only once during [EC](#) creation. Subsequently, each [Host Execution Context](#) is responsible for maintaining its [Stack Pointer \(SP\)](#) across hypercalls. Applications can use different initial [SP](#) or [SEL_{EVNT}](#) values as a means to identify each [Host Execution Context](#) or [Guest Execution Context](#) during concurrent [STARTUP](#) events.

Status:**SUCCESS**

- The hypercall completed successfully.

ABORTED

- A required space of the PD designated by $\text{SPC}_{\text{OBJ}_{\text{CURRENT}}}[\text{pd}]$ is being destructed concurrently.

BAD_CAP

- $\text{SPC}_{\text{OBJ}_{\text{CURRENT}}}[\text{sel}]$ did not refer to a Null Capability (CAP_0).
- $\text{SPC}_{\text{OBJ}_{\text{CURRENT}}}[\text{pd}]$ did not refer to a PD Capability ($\text{CAP}_{\text{OBJ}_{\text{PD}}}$) or that capability had insufficient permissions.

BAD_CPU

- The CPU number is invalid.

BAD_FTR

- Virtual CPUs are not supported by the hardware architecture.

BAD_PAR

- The HVA corresponding to hvp is outside the user-accessible memory range.

MEM_OBJ

- The Protection Domain referred to by $\text{SPC}_{\text{OBJ}_{\text{CURRENT}}}[\text{pd}]$ had insufficient memory resources for object creation.

MEM_CAP

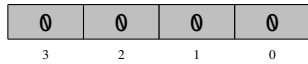
- The Protection Domain to which $\text{SPC}_{\text{OBJ}_{\text{CURRENT}}}$ belongs had insufficient memory resources for capability creation.

5.3.3 Create Scheduling Context

Parameters:

```
status = create_sc (SEL_OBJ sel,          // Created SC
                   SEL_OBJ pd,          // Owner PD
                   SEL_OBJ ec,          // Bound EC
                   SCD   scd)          // Scheduling Context Descriptor
```

Flags:



Description:

Creates a new **Scheduling Context (SC)**.

Prior to the hypercall:

- $SPC_{OBJ_{CURRENT}}[sel]$ must refer to a **Null Capability (CAP_0)**.
- $SPC_{OBJ_{CURRENT}}[pd]$ must refer to a **PD Capability ($CAP_{OBJ_{PD}}$)** with permission SC.
- $SPC_{OBJ_{CURRENT}}[ec]$ must refer to an **EC Capability ($CAP_{OBJ_{EC}}$)** with permission $BIND_{SC}$.
- The owner **PD** designated by $CAP_{OBJ_{PD}}$ and the owner **PD** of the bound **EC** must be the same.

If the hypercall completed successfully:

- A new **Scheduling Context (SC)** has been created.
- The created **SC** is bound to the **EC** referred to by $SPC_{OBJ_{CURRENT}}[ec]$ on the **CPU** of that **EC**, with its scheduling parameters set according to scd .
- The resources for the created **SC** were accounted to the **PD** referred to by $SPC_{OBJ_{CURRENT}}[pd]$.
- $SPC_{OBJ_{CURRENT}}[sel]$ refers to a $CAP_{OBJ_{SC}}$ for the created **SC** with all **defined** permissions.

Status:

SUCCESS

- The hypercall completed successfully.

ABORTED

- The **EC** designated by $SPC_{OBJ_{CURRENT}}[ec]$ is being destructed concurrently.

BAD_CAP

- $SPC_{OBJ_{CURRENT}}[sel]$ did not refer to a **Null Capability (CAP_0)**.
- $SPC_{OBJ_{CURRENT}}[pd]$ did not refer to a **PD Capability ($CAP_{OBJ_{PD}}$)** or that capability had insufficient permissions.
- $SPC_{OBJ_{CURRENT}}[ec]$ did not refer to an **EC Capability ($CAP_{OBJ_{EC}}$)** or that capability had insufficient permissions.
- Binding the **SC** to the **EC** failed, e.g. because the **EC** is a local thread.

BAD_PAR

- At least one **SCD** field in scd was invalid.

MEM_OBJ

- The **Protection Domain** referred to by $SPC_{OBJ_{CURRENT}}[pd]$ had insufficient memory resources for object creation.

MEM_CAP

- The **Protection Domain** to which $SPC_{OBJ_{CURRENT}}$ belongs had insufficient memory resources for capability creation.

5.3.4 Create Portal

Parameters:

```
status = create_pt (SEL_OBJ sel,          // Created PT
                   SEL_OBJ pd,          // Owner PD
                   SEL_OBJ ec,          // Bound EC
                   UINT_PTR ip)         // Instruction Pointer
```

Flags:

0	0	0	0
3	2	1	0

Description:

Creates a new **Portal (PT)**.

Prior to the hypercall:

- $SPC_{OBJ_CURRENT}[sel]$ must refer to a **Null Capability (CAP_0)**.
- $SPC_{OBJ_CURRENT}[pd]$ must refer to a **PD Capability (CAP_{OBJ_PD})** with permission PT.
- $SPC_{OBJ_CURRENT}[ec]$ must refer to an **EC Capability (CAP_{OBJ_EC})** with permission $BIND_{PT}$.
- The owner **PD** designated by CAP_{OBJ_PD} and the owner **PD** of the bound **EC** must be the same.

If the hypercall completed successfully:

- A new **Portal (PT)** has been created.
- The created **PT** is bound to the **EC** referred to by $SPC_{OBJ_CURRENT}[ec]$ on the **CPU** of that **EC**, with its Portal Instruction Pointer (**IP**) set to `ip`, its initial **MTD** set to 0 and its initial **PID** set to 0.
- The resources for the created **PT** were accounted to the **PD** referred to by $SPC_{OBJ_CURRENT}[pd]$.
- $SPC_{OBJ_CURRENT}[sel]$ refers to a CAP_{OBJ_PT} for the created **PT** with all **defined** permissions.

Status:

SUCCESS

- The hypercall completed successfully.

ABORTED

- The **EC** designated by $SPC_{OBJ_CURRENT}[ec]$ is being destructed concurrently.

BAD_CAP

- $SPC_{OBJ_CURRENT}[sel]$ did not refer to a **Null Capability (CAP_0)**.
- $SPC_{OBJ_CURRENT}[pd]$ did not refer to a **PD Capability (CAP_{OBJ_PD})** or that capability had insufficient permissions.
- $SPC_{OBJ_CURRENT}[ec]$ did not refer to an **EC Capability (CAP_{OBJ_EC})** or that capability had insufficient permissions.
- Binding the **PT** to the **EC** failed, e.g. because the **EC** is not a local thread.

MEM_OBJ

- The **Protection Domain** referred to by $SPC_{OBJ_CURRENT}[pd]$ had insufficient memory resources for object creation.

MEM_CAP

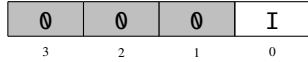
- The **Protection Domain** to which $SPC_{OBJ_CURRENT}$ belongs had insufficient memory resources for capability creation.

5.3.5 Create Semaphore

Parameters:

```
status = create_sm (SEL_OBJ sel,           // Created SM
                   SEL_OBJ pd,           // Owner PD
                   UINT64 val)          // Initial Counter Value / ISD
```

Flags:



Description:

Creates a new **Semaphore (SM)**.

Prior to the hypercall:

- $SPC_{OBJ_{CURRENT}}[sel]$ must refer to a **Null Capability (CAP_0)**.
- If **I=0 (Regular Semaphore)**
 - $SPC_{OBJ_{CURRENT}}[pd]$ must refer to a **PD Capability ($CAP_{OBJ_{PD}}$)** with permission SM.
- If **I=1 (Interrupt Semaphore)**
 - $SPC_{OBJ_{CURRENT}}[pd]$ must refer to a **PD Capability ($CAP_{OBJ_{PD}}$)** with permission DC.
 - The Interrupt Semaphore Descriptor (**ISD**) must be valid.

If the hypercall completed successfully:

- If **I=0 (Regular Semaphore)**
 - A new regular **Semaphore (SM)** has been created.
 - The regular **Semaphore** has an initial counter value of val.
 - The regular **Semaphore** is not associated with any interrupt.
- If **I=1 (Interrupt Semaphore)**
 - A new interrupt **Semaphore (SM)** has been created.
 - The interrupt **Semaphore** has an initial counter value of 0.
 - The interrupt **Semaphore** is permanently associated with the interrupt designated by **ISD**.
- The resources for the created **SM** were accounted to the **PD** referred to by $SPC_{OBJ_{CURRENT}}[pd]$.
- $SPC_{OBJ_{CURRENT}}[sel]$ refers to a $CAP_{OBJ_{SM}}$ for the created **SM** with all **defined** permissions.

Status:

SUCCESS

- The hypercall completed successfully.

ABORTED

- The **PD** designated by $SPC_{OBJ_{CURRENT}}[pd]$ is being destructed concurrently.

BAD_CAP

- $SPC_{OBJ_{CURRENT}}[sel]$ did not refer to a **Null Capability (CAP_0)**.
- $SPC_{OBJ_{CURRENT}}[pd]$ did not refer to a **PD Capability ($CAP_{OBJ_{PD}}$)** or that capability had insufficient permissions.

BAD_PAR

- If **I=1**: The Interrupt Semaphore Descriptor (**ISD**) was invalid.

MEM_OBJ

- The **Protection Domain** referred to by $SPC_{OBJ_{CURRENT}}[pd]$ had insufficient memory resources for object creation.

MEM_CAP

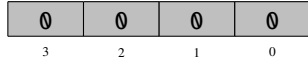
- The **Protection Domain** to which $SPC_{OBJ_{CURRENT}}$ belongs had insufficient memory resources for capability creation.

5.3.6 Create Device Context

Parameters:

```
status = create_dc (SEL_OBJ sel,           // Created DC
                   SEL_OBJ pd,           // Owner PD
                   DTD dtd)              // Device Topology Descriptor
```

Flags:



Description:

Creates a new **Device Context (DC)**.

Prior to the hypercall:

- $SPC_{OBJ_CURRENT}[sel]$ must refer to a **Null Capability (CAP_0)**.
- $SPC_{OBJ_CURRENT}[pd]$ must refer to a **PD Capability (CAP_{OBJ_PD})** with permission DC.

If the hypercall completed successfully:

- A new **Device Context (DC)** has been created.
- The created **DC** has its topology information configured according to **DTD**.
- The resources for the created **DC** were accounted to the **PD** referred to by $SPC_{OBJ_CURRENT}[pd]$.
- $SPC_{OBJ_CURRENT}[sel]$ refers to a CAP_{OBJ_DC} for the created **DC** with all **defined** permissions.

Status:

SUCCESS

- The hypercall completed successfully.

ABORTED

- The **PD** designated by $SPC_{OBJ_CURRENT}[pd]$ is being destructed concurrently.

BAD_CAP

- $SPC_{OBJ_CURRENT}[sel]$ did not refer to a **Null Capability (CAP_0)**.
- $SPC_{OBJ_CURRENT}[pd]$ did not refer to a **PD Capability (CAP_{OBJ_PD})** or that capability had insufficient permissions.

MEM_OBJ

- The **Protection Domain** referred to by $SPC_{OBJ_CURRENT}[pd]$ had insufficient memory resources for object creation.

MEM_CAP

- The **Protection Domain** to which $SPC_{OBJ_CURRENT}$ belongs had insufficient memory resources for capability creation.

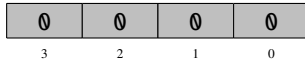
5.4 Object Control

5.4.1 Control Protection Domain

Parameters:

```
status = ctrl_pd (SEL_OBJ src,           // SRC Space
                  SEL_OBJ dst,           // DST Space
                  SEL    ssb,           // SRC Selector Base
                  SEL    dsb,           // DST Selector Base
                  UINT5 ord,           // Order
                  UINT6 pmm,           // Permission Mask
                  MAD     mad)          // Memory Attribute Descriptor
```

Flags:



Description:

Delegates capabilities from the specified selector range in the source space to the specified selector range in the destination space and thereby optionally reduces the permissions for the destination capabilities.

Capability delegation is defined as follows:

1. Capabilities are delegated from source Capability Selector range $\text{src}[\text{ssb}, \text{ssb}+2^{\text{ord}})$.
2. Capabilities are delegated to destination Capability Selector range $\text{dst}[\text{dsb}, \text{dsb}+2^{\text{ord}})$.
3. The permissions of each destination capability are the logical AND of the permissions of the respective source capability and the permission mask pmm , i.e.
 - a) for bits set (1) in pmm , the respective permissions are *inherited* from the source capability.
 - b) for bits clear (0) in pmm , the respective permissions are *removed* for the destination capability.
4. The destination capability is always a Null Capability (CAP_0) in the following cases:
 - a) if the source capability is a Null Capability (CAP_0).
 - b) if the destination capability obtains zero permissions from the source capability.
 - c) if the object that the source capability refers to is being destructed concurrently.

Prior to the hypercall:

- $\text{SPC}_{\text{OBJCURRENT}}[\text{src}]$ must refer to a CAP_{OBJ} for the source space with permission TAKE.
- $\text{SPC}_{\text{OBJCURRENT}}[\text{dst}]$ must refer to a CAP_{OBJ} for the destination space with permission GRANT.
- Capability Selectors ssb and dsb must be order-aligned: $\text{ssb} \equiv 0 \pmod{2^{\text{ord}}}$ and $\text{dsb} \equiv 0 \pmod{2^{\text{ord}}}$.
- Capability Selectors ssb and dsb must be equal if src refers to a PIO Space or an MSR Space.

If the hypercall completed successfully:

- If both src and dst refer to Object Spaces:
 - All CAP_{OBJ} (or CAP_0) from the source selector range were delegated to the destination selector range. Any pre-existing CAP_{OBJ} in the destination selector range were revoked.
 - The parameter mad was ignored.
- If src refers to a Host Space and dst refers to a Host Space or Guest Space or DMA Space:
 - All CAP_{MEM} (or CAP_0) from the source selector range were delegated to the destination selector range. Any pre-existing CAP_{MEM} in the destination selector range were revoked.
 - If src refers to the NOVA Host Space, then source SEL_{HST} are **physical** page numbers and the memory attributes of each destination CAP_{MEM} were *set* according to mad .

Otherwise, source SEL_{HST} are **virtual** page numbers and the memory attributes of each destination CAP_{MEM} were *inherited* from the respective source CAP_{MEM} , i.e. the parameter mad was ignored.

- If both src and dst refer to **PIO Spaces**:
 - All CAP_{PIO} (or CAP_0) from the source selector range were delegated to the destination selector range. Any pre-existing CAP_{PIO} in the destination selector range were revoked.
 - The parameter mad was ignored.
- If both src and dst refer to **MSR Spaces**:
 - All CAP_{MSR} (or CAP_0) from the source selector range were delegated to the destination selector range. Any pre-existing CAP_{MSR} in the destination selector range were revoked.
 - The parameter mad was ignored.
- The resources for storing the granted capabilities were accounted to the **PD** to which the space referred to by $SPC_{OBJ_{CURRENT}}[dst]$ belongs.

Status:

SUCCESS

- The hypercall completed successfully.

BAD_CAP

- $SPC_{OBJ_{CURRENT}}[src]$ or $SPC_{OBJ_{CURRENT}}[dst]$ did not refer to compatible space capabilities or had insufficient permissions.

BAD_PAR

- **Capability Selector** ssb or dsb was not order-aligned.
- **Capability Selector** ssb+2^{ord} or dsb+2^{ord} was larger than the number of selectors.
- If src refers to a **PIO Space** or an **MSR Space**: **Capability Selectors** ssb and dsb were not equal.
- If src refers to the **NOVA Host Space**: At least one **MAD** field in mad was invalid.

MEM_CAP

- The **PD** to which the space referred to by $SPC_{OBJ_{CURRENT}}[dst]$ belongs had insufficient memory resources for allocating the storage required for granting all destination capabilities. This constitutes a partial failure of the operation, because all destination capabilities up to the first allocation failure have been granted.

5.4.2 Control Execution Context

Parameters:

```
status = ctrl_ec (SEL_OBJ ec)           // Execution Context
```

Flags:

0	0	0	S
3	2	1	0

Description:

Prior to the hypercall:

- $\text{SPC}_{\text{OBJ}_{\text{CURRENT}}}[\text{ec}]$ must refer to an **EC Capability** ($\text{CAP}_{\text{OBJ}_{\text{EC}}}$) with permission CTRL.

If the hypercall completed successfully:

- The **EC** referred to by $\text{SPC}_{\text{OBJ}_{\text{CURRENT}}}[\text{ec}]$ has been forced to enter the microhypervisor. It will generate a recall exception prior to its next exit from the microhypervisor and will traverse through the respective Event Portal (Arm, x86).
- If **S=0 (Weak Recall)**:
 - The hypercall returns as soon as the recall exception has been *pending*, i.e. the **EC** may not have entered the microhypervisor yet.
- If **S=1 (Strong Recall)**:
 - The hypercall returns as soon as the recall exception has been *observed*, i.e the **EC** will have entered the microhypervisor.

Status:

SUCCESS

- The hypercall completed successfully.

BAD_CAP

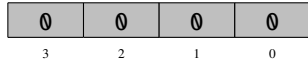
- $\text{SPC}_{\text{OBJ}_{\text{CURRENT}}}[\text{ec}]$ did not refer to an **EC Capability** ($\text{CAP}_{\text{OBJ}_{\text{EC}}}$) or that capability had insufficient permissions.

5.4.3 Control Scheduling Context

Parameters:

```
status = ctrl_sc (SEL_OBJ sc,          // Scheduling Context
                  UINT64 &stc)         // Total Consumed Execution Time
```

Flags:



Description:

Prior to the hypercall:

- `SPC_OBJ_CURRENT[sc]` must refer to an **SC Capability** (`CAP_OBJ_SC`) with permission CTRL.

If the hypercall completed successfully:

- The microhypervisor has returned the total consumed execution time as **System Time Counter (STC)** value for the **SC** referred to by `SPC_OBJ_CURRENT[sc]`.

Status:

SUCCESS

- The hypercall completed successfully.

BAD_CAP

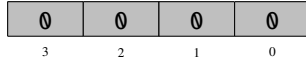
- `SPC_OBJ_CURRENT[sc]` did not refer to an **SC Capability** (`CAP_OBJ_SC`) or that capability had insufficient permissions.

5.4.4 Control Portal

Parameters:

```
status = ctrl_pt (SEL_OBJ pt,           // Portal
                  UINT_PTR pid,         // Portal Identifier
                  MTD mtd)              // Message Transfer Descriptor
```

Flags:



Description:

Prior to the hypercall:

- $SPC_{OBJ_CURRENT}[pt]$ must refer to a **PT Capability** (CAP_{OBJ_PT}) with permission CTRL.

If the hypercall completed successfully:

- The microhypervisor has set the Portal Identifier (**PID**) to `pid` and the **Message Transfer Descriptor** (**MTD**) to `mtd` for the **Portal** referred to by $SPC_{OBJ_CURRENT}[pt]$.
- Subsequent portal traversals will use the new **MTD** and return the new **PID**.

Status:

SUCCESS

- The hypercall completed successfully.

BAD_CAP

- $SPC_{OBJ_CURRENT}[pt]$ did not refer to a **PT Capability** (CAP_{OBJ_PT}) or that capability had insufficient permissions.

5.4.5 Control Semaphore

Parameters:

```
status = ctrl_sm (SEL_OBJ sm,           // Semaphore
                  UINT64 stc)           // Absolute Timeout
```

Flags:

0	0	Z	D
3	2	1	0

Description:

Prior to the hypercall:

- If **D=0 (Semaphore Up)**:
 - $SPC_{OBJ_{CURRENT}}[sm]$ must refer to an **SM Capability** ($CAP_{OBJ_{SM}}$) with permission $CTRL_{UP}$.
- If **D=1 (Semaphore Down)**:
 - $SPC_{OBJ_{CURRENT}}[sm]$ must refer to an **SM Capability** ($CAP_{OBJ_{SM}}$) with permission $CTRL_{DN}$.

If the hypercall completed successfully:

- If **D=0 (Semaphore Up)**:
 - If there were **ECs** blocked on the semaphore, then the microhypervisor has released one of those blocked **ECs**. Otherwise, the microhypervisor has incremented the semaphore counter. The timeout value and the Z-flag were ignored.
- If **D=1 (Semaphore Down)**:
 - If the semaphore counter was larger than zero, then the microhypervisor has decremented the semaphore counter (**Z=0**) or set it to zero (**Z=1**). Otherwise, the microhypervisor has blocked $EC_{CURRENT}$ on the semaphore. If the timeout value was non-zero, $EC_{CURRENT}$ unblocks with a timeout status when the **System Time Counter (STC)** reaches or exceeds the specified value.

Blocking and releasing of **ECs** on a semaphore uses the FIFO queueing discipline.

Status:

SUCCESS

- The hypercall completed successfully.

TIMEOUT

- If **D=1**: Down operation aborted when the timeout triggered.

OVRFLOW

- If **D=0**: Up operation aborted because the semaphore counter would overflow.

BAD_CAP

- $SPC_{OBJ_{CURRENT}}[sm]$ did not refer to an **SM Capability** ($CAP_{OBJ_{SM}}$) or that capability had insufficient permissions.

BAD_CPU

- If **D=1** on an interrupt semaphore: Attempt to wait for the interrupt on a different **CPU** than the **CPU** to which that interrupt has been routed via `assign_int`.

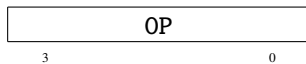
5.5 Platform Management

5.5.1 Control Hardware

Parameters:

```
status = ctrl_hw (UINT desc)           // Descriptor
```

Flags:



Description:

Modifies the platform hardware configuration or power management state.

Prior to the hypercall:

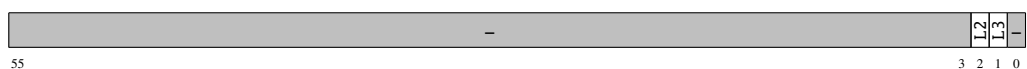
- The hypercall must be invoked by an [Execution Context](#) in the [Root Protection Domain](#) (PD_{ROOT}).
- If **OP=0 (S-State Transition)**:
 - The descriptor desc uses the following encoding:



- The value S designates the platform-wide reset or sleep state that shall be entered. The values A and B are the first two bytes of the respective $\backslash_Sx$ package in the [ACPI](#) root namespace as follows:

S	A	B	Type	Description
0x0	0x0	0x0	Reset	Platform Reset
0x1	$\backslash_S1[0]$	$\backslash_S1[1]$	Shallow	S1: Stop Grant
0x2	$\backslash_S2[0]$	$\backslash_S2[1]$		S2: Power-On Suspend
0x3	$\backslash_S3[0]$	$\backslash_S3[1]$		S3: Suspend to RAM
0x4	$\backslash_S4[0]$	$\backslash_S4[1]$	Deep	S4: Suspend to Disk
0x5	$\backslash_S5[0]$	$\backslash_S5[1]$		S5: Soft Off

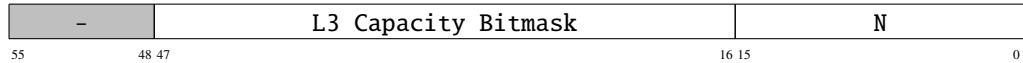
- The caller is responsible for invoking the necessary pre-sleep [ACPI](#) methods, for transitioning platform devices into a suitable Dx sleep state, and for programming wakeup events.
- If **OP=4 (QOS Configuration)**:
 - The descriptor desc uses the following encoding:



- Only Code and Data Prioritization ([CDP](#)) settings supported by the ambient [CPU](#) are valid.
 - * The L3 bit disables (0) or enables (1) CDP_{L3} on the ambient [CPU](#).
 - * The L2 bit disables (0) or enables (1) CDP_{L2} on the ambient [CPU](#).
- [CAT/CDP](#) or [MBA](#) settings cannot be configured for a [CPU](#) until a valid [QOS](#) configuration has been established for that [CPU](#). Subsequently, that [QOS](#) configuration cannot be changed anymore.

- If **OP=5 (CAT/CDP L3 Capacity Bitmask)**:

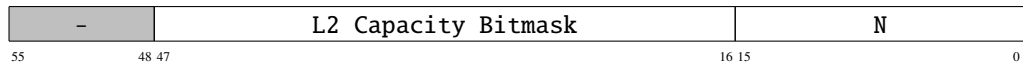
- The descriptor desc uses the following encoding:



- N designates the CPU-local Class Of Service (COS) that shall be configured. Only COS below COS_{L3} of the ambient CPU are valid.
 - * If CDP_{L3} is disabled on the ambient CPU:
 - To configure the CAT_{L3} Capacity Bitmask for a COS, use N=COS.
 - * If CDP_{L3} is enabled on the ambient CPU:
 - To configure the CDP_{L3-Data} Capacity Bitmask for a COS, use N=(COS<<1).
 - To configure the CDP_{L3-Code} Capacity Bitmask for a COS, use N=(COS<<1)+1.
- For the L3 Capacity Bitmask, all (and only) contiguous combinations of 1-bits up to the highest capacity bit supported by the ambient CPU are valid.

- If **OP=6 (CAT/CDP L2 Capacity Bitmask)**:

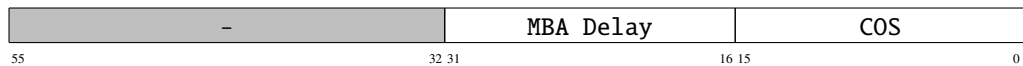
- The descriptor desc uses the following encoding:



- N designates the CPU-local Class Of Service (COS) that shall be configured. Only COS below COS_{L2} of the ambient CPU are valid.
 - * If CDP_{L2} is disabled on the ambient CPU:
 - To configure the CAT_{L2} Capacity Bitmask for a COS, use N=COS.
 - * If CDP_{L2} is enabled on the ambient CPU:
 - To configure the CDP_{L2-Data} Capacity Bitmask for a COS, use N=(COS<<1).
 - To configure the CDP_{L2-Code} Capacity Bitmask for a COS, use N=(COS<<1)+1.
- For the L2 Capacity Bitmask, all (and only) contiguous combinations of 1-bits up to the highest capacity bit supported by the ambient CPU are valid.

- If **OP=7 (MBA Delay)**:

- The descriptor desc uses the following encoding:



- COS designates the CPU-local Class Of Service (COS) that shall be configured. Only COS below COS_{MB} of the ambient CPU are valid.
- For the MBA Delay, only values up to the highest delay supported by the ambient CPU are valid.

If the hypercall completed successfully:

- If **OP=0 (S-State Transition)**:

- The platform resets or enters the specified ACPI sleep state.
- For reset or deep sleep states, the hypercall does not return.
- For shallow sleep states, the hypercall returns upon a wakeup event. The caller is responsible for invoking the necessary post-sleep ACPI methods and for transitioning platform devices back into the D0 working state.

- If **OP=4 (QOS Configuration)**:

- The ambient CPU uses and locks down the QOS configuration.

- If **OP=5 (CAT/CDP L3 Capacity Bitmask)**:

- The ambient CPU uses the L3 Capacity Bitmask for the designated COS.

- If **OP=6 (CAT/CDP L2 Capacity Bitmask)**:

- The ambient CPU uses the L2 Capacity Bitmask for the designated COS.
- If OP=7 (MBA Delay):
 - The ambient CPU uses the MBA Delay for the designated COS.

Status:

SUCCESS

- The hypercall completed successfully.

BAD_HYP

- The hypercall was not invoked from the Root Protection Domain (PD_{ROOT}).

BAD_FTR

- The requested feature is not supported by the platform.

BAD_PAR

- If OP=4: The QOS configuration is invalid.
- If OP=5/6/7: The COS, CAT/CDP capacity bitmask or MBA delay is invalid.
- Otherwise: The requested operation (OP) is invalid.

ABORTED

- If OP=0: A concurrent power management transition prevailed.
- If OP=4: A QOS configuration has already been established for the ambient CPU.
- If OP=5/6/7: A QOS configuration has not yet been established for the ambient CPU.

5.5.2 Assign Device

Parameters:

```
status = assign_dev (SEL_OBJ dc,          // DMA Source Device
                    SEL_OBJ dma)         // DMA Space
```

Flags:

0	0	0	0
3	2	1	0

Description:

Assigns a DMA-capable device to the specified **DMA Space** (SPC_{DMA}).

Prior to the hypercall:

- $SPC_{OBJ_{CURRENT}}[dc]$ must refer to a **DC Capability** ($CAP_{OBJ_{DC}}$) with permission $ASSIGN_{DEV}$.
- $SPC_{OBJ_{CURRENT}}[dma]$ must refer to a **DMA Space Capability** ($CAP_{OBJ_{DMA}}$) with permission $ASSIGN$.

If the hypercall completed successfully:

- The device designated by $SPC_{OBJ_{CURRENT}}[dc]$ has been assigned to the **DMA Space** designated by $SPC_{OBJ_{CURRENT}}[dma]$.
- DMA transactions of that device will be managed by the **SMMU** and its resources that were encoded in the **Device Context** during its creation. Prior users of those **SMMU** resources have been unconfigured.

Status:

SUCCESS

- The hypercall completed successfully.

TIMEOUT

- An **SMMU** command timed out at the hardware level.

ABORTED

- A concurrent **assign_dev** for the designated device prevailed.

BAD_CAP

- $SPC_{OBJ_{CURRENT}}[dc]$ did not refer to a **DC Capability** ($CAP_{OBJ_{DC}}$) or that capability had insufficient permissions.
- $SPC_{OBJ_{CURRENT}}[dma]$ did not refer to a **DMA Space Capability** ($CAP_{OBJ_{DMA}}$) or that capability had insufficient permissions.

BAD_DEV

- The **SMMU** configured in the **Device Context** is invalid.

MEM_OBJ

- A translation table for the **DMA Space** or a mapping table for the **SMMU** could not be allocated.

5.5.3 Assign Interrupt

Parameters:

```
status = assign_int (SEL_OBJ sm,          // Interrupt Semaphore (GSI)
                    SEL_OBJ dc,          // MSI Source Device
                    UINT_4  cfg,         // Interrupt Configuration
                    UINT_16 cpu,         // Destination CPU Number
                    UINT_16 idx,         // Table Index
                    UINT_PTR &msi_addr,  // OUT: MSI Address
                    UINT_PTR &msi_data) // OUT: MSI Data
```

Flags:

0	0	0	A
3	2	1	0

Description:

Attaches or Detaches an Interrupt **Semaphore** in a semaphore table slot and configures the associated **GSI**.

Prior to the hypercall:

- $SPC_{OBJ_{CURRENT}}[sm]$ must refer to an **SM Capability** ($CAP_{OBJ_{SM}}$) for an Interrupt **Semaphore** with permission **ASSIGN** and thereby designates the **GSI**.
- The destination **CPU** must be valid ($cpu \leq CPU_{MAX}$).
- The **Semaphore** (**SM**) table slot is determined as follows:
 - **Arm**: The **SM** table is shared by all **CPUs** and indexed by **GSI**.
 - **x86**: The **SM** table is individual for each **CPU**, designated by **cpu** and indexed by **idx**.
- If **A=0** (**Detach**)
 - The **SM** table slot must be attached to the Interrupt **Semaphore** designated by $SPC_{OBJ_{CURRENT}}[sm]$.
 - $SPC_{OBJ_{CURRENT}}[dc]$ must refer to a **Null Capability** (CAP_0).
- If **A=1** (**Attach**)
 - The **SM** table slot must be attached to the Interrupt **Semaphore** designated by $SPC_{OBJ_{CURRENT}}[sm]$ or must be detached.
 - If the **GSI** type is **PIN**
 - * $SPC_{OBJ_{CURRENT}}[dc]$ must refer to a **Null Capability** (CAP_0).
 - If the **GSI** type is **MSI**
 - * $SPC_{OBJ_{CURRENT}}[dc]$ must refer to a **DC Capability** ($CAP_{OBJ_{DC}}$) with permission $ASSIGN_{INT}$.

If the hypercall completed successfully:

- If **A=0** (**Detach**)
 - The Interrupt **Semaphore** designated by $SPC_{OBJ_{CURRENT}}[sm]$ has been detached from the **SM** table slot and will no longer signal interrupt assertions arriving at that **SM** table slot.
- If **A=1** (**Attach**)
 - The Interrupt **Semaphore** designated by $SPC_{OBJ_{CURRENT}}[sm]$ has been attached to the **SM** table slot and will signal interrupt assertions arriving at that **SM** table slot.
 - The **GSI** associated with that Interrupt **Semaphore** has been routed to the designated **CPU** and **SM** table slot and configured according to **cfg**, which uses the following encoding:

<table><tr><td>O</td><td>P</td><td>T</td><td>M</td></tr><tr><td>3</td><td>2</td><td>1</td><td>0</td></tr></table>	O	P	T	M	3	2	1	0	Mask	0 = Unmasked	1 = Masked [†]
O	P	T	M								
3	2	1	0								
	Trigger	0 = Edge-Triggered	1 = Level-Triggered [†]								
	Polarity	0 = Active-High	1 = Active-Low [†]								
	Ownership	0 = Host-Owned	1 = Guest-Owned [‡]								

[†]Only valid for GSI type PIN

[‡]Only valid for the Arm architecture

- If the **GSI** type is **PIN**
 - * Any device that is connected to that interrupt pin is able to generate the interrupt.
 - * The returned `msi_{addr, data}` values are 0 and meaningless.
- If the **GSI** type is **MSI**
 - * **Arm:** **GITS** translates the device-specific EventID `idx` to the **MSI**.
 - * Only the device designated by `SPC_OBJ_CURRENT[dc]` is authorized to generate the **MSI**.
 - * The device driver must program the returned `msi_{addr, data}` values into the **MSI** registers of that device to ensure proper interrupt operation.

Prior to the first invocation of `assign_int` for a **GSI**, the state of that interrupt is as follows:

- The interrupt is masked.
- The configuration (trigger, polarity, ownership) is undefined.
- The source device, destination **CPU** and semaphore table index are undefined.

Status:

SUCCESS

- The hypercall completed successfully.

BAD_CPU

- The specified CPU number was invalid.

BAD_CAP

- `SPC_OBJ_CURRENT[sm]` did not refer to an **SM Capability** (`CAP_OBJ_SM`) for an Interrupt **Semaphore** or that capability had insufficient permissions.
- If Attach (PIN) or Detach: `SPC_OBJ_CURRENT[dc]` did not refer to a **Null Capability** (`CAP0`).
- If Attach (MSI): `SPC_OBJ_CURRENT[dc]` did not refer to a **DC Capability** (`CAP_OBJ_DC`) or that capability had insufficient permissions.

BAD_DEV

- The **Device Context (DC)** configuration was invalid.

BAD_PAR

- The semaphore table index was invalid.
- The interrupt configuration was invalid.

ABORTED

- A concurrent operation destroyed the designated interrupt semaphore.
- The semaphore table slot was not in the required state.

6 Booting

6.1 NOVA Microhypervisor

6.1.1 NOVA Image

The bootloader must place all loadable (PT_LOAD) program segments of the [NOVA](#) microhypervisor into physical memory (RAM) according to the physical addresses (`p_paddr`) and memory sizes (`p_memsz`) defined in the [NOVA](#) microhypervisor [ELF](#) executable. The following is an example:

```
readelf -l nova.elf
```

Elf file type is EXEC (Executable file)

Entry point 0x404000

There are 2 program headers, starting at offset 64

Program Headers:

Type	Offset FileSiz	VirtAddr MemSiz	PhysAddr Flags Align
LOAD	0x00000000000000e8 0x0000000000000091c	0x00000000000404000 0x0000000000000091c	0x00000000000404000 RW 0x8
LOAD	0x0000000000000191c 0x00000000000014d4c	0xffffffff80000491c 0x000000000000ffb6e4	0x0000000000040491c RW 0x1000

If the physical memory range defined in the [ELF](#) executable is unsuitable for a particular platform, the bootloader may shift all loadable program segments lower or higher in physical memory, by applying an offset, subject to the following constraints:

- The same offset must be applied to each loadable program segment and to the entry point.
- The offset must comply with the load range and image alignment specified in the Multiboot2 [6] relocatable header tag that is embedded in the [NOVA](#) microhypervisor [ELF](#) executable.
- The entire physical memory region occupied by the [NOVA](#) microhypervisor must be RAM.

After loading the [NOVA](#) microhypervisor into physical memory, the bootloader must invoke the entry point of the [ELF](#) executable with architecture-specific preconditions ([Arm](#), [x86](#)).

6.1.2 NOVA Integrity Measurement

The Reference Integrity Measurement ([RIM](#)) of the [NOVA](#) microhypervisor is printed during **make install**.

On hardware platforms with support for establishing a Dynamic Root of Trust for Measurement ([DRTM](#)), the [NOVA](#) microhypervisor initiates a measured launch of itself. The Core Root of Trust for Measurement ([CRTM](#)) extends a Launch Integrity Measurement ([LIM](#)) of the attestable region of the [NOVA](#) microhypervisor into [PCR 17](#) of the Trusted Platform Module ([TPM](#)). The Launch Integrity Measurement is

- sensitive to ([LIM](#) ≠ [RIM](#))
 - any changes in the attestable region
 - the bootloader passing command-line parameters to [NOVA](#)
- insensitive to ([LIM](#) = [RIM](#))
 - the hardware platform and [NOVA](#) patching its code to adapt to available hardware features
 - the physical memory range into which [NOVA](#) has been loaded
 - the memory map and other software modules that have been loaded

6.1.3 NOVA Spaces

6.1.3.1 NOVA Object Space

The **NOVA Object Space** ($SPC_{OBJ_{NOVA}}$) contains the following **CAP_{OBJ}**:

SEL_{OBJ}	Capability Type	Capability Resource	Capability Permissions
$2^{SBW_{OBJ}-1}$	CAP_{OBJ_{SM}}}	Console Semaphore	All defined permissions
$2^{SBW_{OBJ}-2}$	CAP_{OBJ_{OBJ}}}	NOVA Object Space	TAKE
$2^{SBW_{OBJ}-3}$	CAP_{OBJ_{HST}}}	NOVA Host Space	TAKE
$2^{SBW_{OBJ}-4}$	CAP_{OBJ_{PIO}}}	NOVA PIO Space [†]	TAKE
$2^{SBW_{OBJ}-5}$	CAP_{OBJ_{MSR}}}	NOVA MSR Space [†]	TAKE
$2^{SBW_{OBJ}-6}$	CAP_{OBJ_{OBJ}}}	Root Object Space	All defined permissions
$2^{SBW_{OBJ}-7}$	CAP_{OBJ_{HST}}}	Root Host Space	All defined permissions
$2^{SBW_{OBJ}-8}$	CAP_{OBJ_{PIO}}}	Root PIO Space [†]	All defined permissions
$0 \dots CPU_{MAX}$	CAP_{OBJ_{SC}}}	Idle Scheduling Contexts	CTRL

Using $SPC_{OBJ_{NOVA}}$ as source space for the **ctrl_pd** hypercall facilitates the delegation of **CAP_{OBJ}** from the **NOVA Object Space** to another **Object Space**.

6.1.3.2 NOVA Host Space

In the **NOVA Host Space** ($SPC_{HST_{NOVA}}$), **SEL_{HST} N** refers to

- **CAP₀** for memory protected by the NOVA microhypervisor (**Arm**, **x86**).
- **CAP_{MEM}** for the 4 KiB page frame at **physical** address $N \ll 12$ with all **defined** permissions otherwise.

Using $SPC_{HST_{NOVA}}$ as source space for the **ctrl_pd** hypercall facilitates the delegation of **CAP_{MEM}** from the **NOVA Host Space** to another **Host Space**, **Guest Space** or **DMA Space**.

6.1.3.3 NOVA PIO Space

In the **NOVA PIO Space** ($SPC_{PIO_{NOVA}}$), **SEL_{PIO} N** refers to

- **CAP₀** for an I/O port protected by the NOVA microhypervisor (**x86**).
- **CAP_{PIO}** for the I/O port number N with certain permissions otherwise (**x86**).

Using $SPC_{PIO_{NOVA}}$ as source space for the **ctrl_pd** hypercall facilitates the delegation of **CAP_{PIO}** from the **NOVA PIO Space** to another **PIO Space**.

6.1.3.4 NOVA MSR Space

In the **NOVA MSR Space** ($SPC_{MSR_{NOVA}}$), **SEL_{MSR} N** refers to

- **CAP₀** for an **MSR** protected by the NOVA microhypervisor (**x86**).
- **CAP_{MSR}** for the **MSR** number N with certain permissions otherwise (**x86**).

Using $SPC_{MSR_{NOVA}}$ as source space for the **ctrl_pd** hypercall facilitates the delegation of **CAP_{MSR}** from the **NOVA MSR Space** to another **MSR Space**.

[†]Only on architectures, where this type of space **exists**, otherwise **Null Capability** (**CAP₀**).

6.2 Root Protection Domain

After the [NOVA](#) microhypervisor has initialized the system, it creates the following initial kernel objects:

- [Root Protection Domain](#) (PD_{ROOT})
 - with [Root Object Space](#) ($SPC_{OBJ_{ROOT}}$)
 - with [Root Host Space](#) ($SPC_{HST_{ROOT}}$)
 - with [Root PIO Space](#) ($SPC_{PIO_{ROOT}}$)[†]
- [Root Execution Context](#) (EC_{ROOT}) on CPU_{BSP}
 - bound to $SPC_{OBJ_{ROOT}}$, $SPC_{HST_{ROOT}}$, $SPC_{PIO_{ROOT}}$ [†]
 - with $SEL_{EVT} = 0$
- [Root Scheduling Context](#) (SC_{ROOT}) on CPU_{BSP}
 - bound to EC_{ROOT}
 - with $COS = 0$
 - with Priority = highest priority
 - with Budget = 1000 ms

The [Root Protection Domain](#) is responsible for bootstrapping the other components of the user-mode framework by creating additional kernel objects, loading additional images, assigning resources, etc.

6.2.1 Root Image

The image of the [Root Protection Domain](#) (PD_{ROOT}) must be a valid [ELF](#) executable (ET_EXEC) that has been compiled for the respective hardware architecture and

- linked such that $p_filesz = p_memsz$
- loaded such that $p_vaddr \equiv LOAD_ADDR^* + p_offset \pmod{PAGE_SIZE}$

holds for each loadable (PT_LOAD) program segment. These constraints ensure that the [NOVA](#) microhypervisor can map all program segments directly from physical into virtual memory without any additional memory allocation or copying. The following is an example:

```
readelf -l root.elf
```

Elf file type is EXEC (Executable file)

Entry point 0x10000120

There are 2 program headers, starting at offset 64

Program Headers:

Type	Offset	VirtAddr	PhysAddr
	FileSiz	MemSiz	Flags Align
LOAD	0x0000000000000000	0x0000000010000000	0x0000000010000000
	0x00000000000000a75	0x00000000000000a75	R E 0x1000
LOAD	0x00000000000001000	0x0000000010001000	0x0000000010001000
	0x0000000000000f004	0x0000000000000f004	RW 0x1000

[†]Only on architectures, where this type of space [exists](#).

*The address in physical memory at which the bootloader has placed the ELF image.

6.2.2 Root Integrity Measurement

On hardware platforms with support for establishing a Dynamic Root of Trust for Measurement (DRTM), the NOVA microhypervisor extends a Launch Integrity Measurement (LIM) of the attestable region of the Root Protection Domain into PCR 19 of the Trusted Platform Module (TPM).

The attestable region is defined as the entire contents of the first loadable (PT_LOAD) program segment of the PD_{ROOT} ELF image that is readable and/or executable, but not writable.

The attestable region of the Root Protection Domain should cover:

- the ELF header – to include the entry point of PD_{ROOT} in the measurement
- the ELF program headers – to include the virtual-memory layout of PD_{ROOT} in the measurement
- the code and read-only data sections of PD_{ROOT} – to reflect their integrity

6.2.3 Root Spaces

Prior to the Root Execution Context (EC_{ROOT}) invoking the entry point of the Root Protection Domain (PD_{ROOT}) ELF image, the NOVA microhypervisor sets up the spaces for PD_{ROOT} as described in the following subsections.

6.2.3.1 Root Object Space

The Root Object Space (SPC_{OBJ_{ROOT}}) contains the following initial CAP_{OBJ}:

SEL _{OBJ}	Capability Type	Capability Resource	Capability Permissions
2 ^{SBW_{OBJ}-1}	CAP _{OBJ_{OBJ}}	NOVA Object Space	TAKE
2 ^{SBW_{OBJ}-2}	CAP _{OBJ_{OBJ}}	Root Object Space	All defined permissions
2 ^{SBW_{OBJ}-3}	CAP _{OBJ_{PD}}	Root Protection Domain	All defined permissions
2 ^{SBW_{OBJ}-4}	CAP _{OBJ_{EC}}	Root Execution Context	All defined permissions
2 ^{SBW_{OBJ}-5}	CAP _{OBJ_{SC}}	Root Scheduling Context	All defined permissions

All other SEL_{OBJ} in SPC_{OBJ_{ROOT}} initially refer to a Null Capability (CAP₀).

6.2.3.2 Root Host Space

ELF Program Segments

The microhypervisor maps the Root Protection Domain (PD_{ROOT}) into the Root Host Space according to the virtual addresses (p_vaddr), memory sizes (p_memsz) and page attributes (p_flags) of all loadable (PT_LOAD) program segments defined in the PD_{ROOT} ELF image.

Hypervisor Information Page

The microhypervisor maps the Hypervisor Information Page read-only into the Root Host Space 4 KiB below the end of user-accessible virtual memory. The virtual address of the HIP is passed to EC_{ROOT} at the entry point (Arm, x86).

UTCB

The microhypervisor maps the User Thread Control Block of EC_{ROOT} into the Root Host Space 4 KiB below the address of the Hypervisor Information Page.

All other SEL_{HST} in SPC_{HST_{ROOT}} initially refer to a Null Capability (CAP₀).

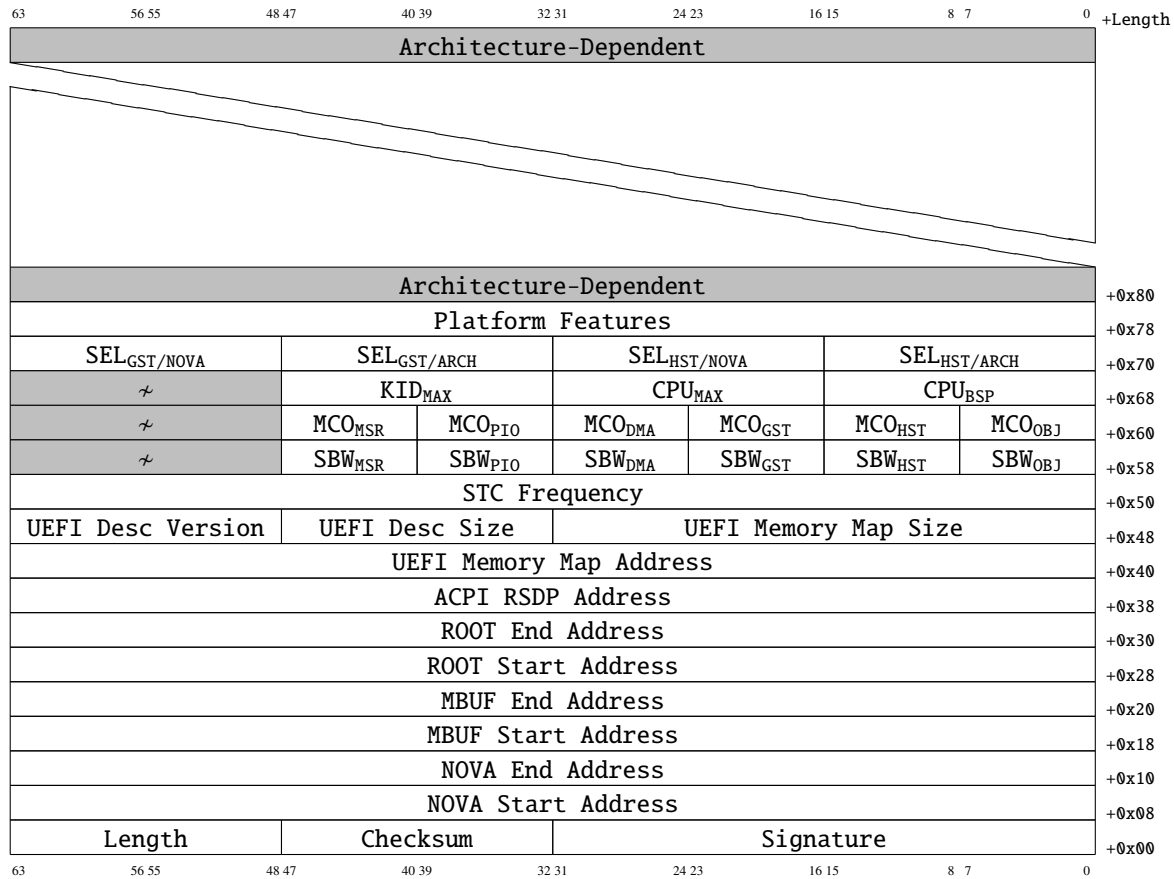
6.2.3.3 Root PIO Space

All SEL_{PIO} in SPC_{PIO_{ROOT}} initially refer to a Null Capability (CAP₀).

6.3 Hypervisor Information Page

The [Hypervisor Information Page \(HIP\)](#) is mapped in the [Host Space](#) of the [Root Protection Domain](#) and the initial Stack Pointer ([SP](#)) of the [Root Execution Context](#) points to the [HIP](#). Because the [HIP](#) is allocated and owned by the microhypervisor, it cannot be delegated using `ctrl_pd`.

The [HIP](#) conveys information about the platform and configuration and has the following layout:



All HIP fields are unsigned little-endian values, unless stated otherwise, and have the following meaning:

Signature

The value `0x41564f4e` identifies the [NOVA](#) microhypervisor.

Checksum

The checksum is valid if 16bit-wise addition of the entire [HIP](#) contents produces a value of `0`.

Length

Length of the entire [HIP](#) in bytes.

NOVA Start/End Address

Physical start and end address of the [NOVA](#) microhypervisor image.

MBUF Start/End Address

Physical start and end address of the memory buffer console region (see [C.1](#)).

ROOT Start/End Address

Physical start and end address of the root protection domain image.

ACPI RSDP Address

Physical address of the [ACPI](#) [7] Root System Description Pointer (`0xffffffffffffffff` if not present).

UEFI Memory Map Address

Physical address of the [UEFI](#) [8] Memory Map (`0xffffffffffffffff` if not present).

UEFI Memory Map Size

Total size of the [UEFI Memory Map](#) (0 if not present).

UEFI Desc Size

[UEFI Memory Descriptor Size](#) (0 if not present).

UEFI Desc Version

[UEFI Memory Descriptor Version](#) (0 if not present).

STC Frequency

Frequency of the [System Time Counter \(STC\)](#) in Hz.

SBW_{OBJ}

Selector bit width: [Object Capability Selector \(SEL_{OBJ}\)](#)

SBW_{HST}

Selector bit width: [Host Capability Selector \(SEL_{HST}\)](#)

SBW_{GST}

Selector bit width: [Guest Capability Selector \(SEL_{GST}\)](#)

SBW_{DMA}

Selector bit width: [DMA Capability Selector \(SEL_{DMA}\)](#)

SBW_{PIO}

Selector bit width: [PIO Capability Selector \(SEL_{PIO}\)](#)

SBW_{MSR}

Selector bit width: [MSR Capability Selector \(SEL_{MSR}\)](#)

MCO_{OBJ}

Maximum contiguous order that avoids partial failures during capability updates in [SPC_{OBJ}](#).

MCO_{HST}

Maximum contiguous order that avoids partial failures during capability updates in [SPC_{HST}](#).

MCO_{GST}

Maximum contiguous order that avoids partial failures during capability updates in [SPC_{GST}](#).

MCO_{DMA}

Maximum contiguous order that avoids partial failures during capability updates in [SPC_{DMA}](#).

MCO_{PIO}

Maximum contiguous order that avoids partial failures during capability updates in [SPC_{PIO}](#).

MCO_{MSR}

Maximum contiguous order that avoids partial failures during capability updates in [SPC_{MSR}](#).

CPU_{BSP}

Bootstrap Processor. The [CPU](#) number on which [EC_{ROOT}](#) and [SC_{ROOT}](#) have been created.

CPU_{MAX}

Maximum [CPU](#) number. The total number of [CPU](#)s is CPU_{MAX} + 1.

KID_{MAX}

Maximum [KID](#) number. The total number of [KID](#)s is KID_{MAX} + 1.

SEL_{HST/ARCH}

Number of [Capability Selectors](#) required for handling architectural host events. ([Arm, x86](#))

SEL_{HST/NOVA}

Number of additional [Capability Selectors](#) required for handling microhypervisor host events. ([Arm, x86](#))

SEL_{GST/ARCH}

Number of [Capability Selectors](#) required for handling architectural guest events. ([Arm, x86](#))

SEL_{GST/NOVA}

Number of additional [Capability Selectors](#) required for handling microhypervisor guest events. ([Arm, x86](#))

Platform Features

Supported platform features. ([Arm, x86](#))

Architecture-Dependent

Architecture-dependent part. ([Arm](#), [x86](#))

Part IV

Application Binary Interface

7 ABI aarch64

7.1 Boot State

7.1.1 NOVA Microhypervisor

The bootloader must set up the [CPU](#) register state according to one of the launch types listed below when it transfers control to the [NOVA](#) microhypervisor entry point. Furthermore, the following preconditions must be satisfied:

- The [CPU](#) must execute in EL2 (hypervisor mode) or in EL3 (monitor mode).
- Paging ([MMU](#)) must either be disabled (`SCTLR_ELx.M=0`) or the page tables must provide an identity (1:1) mapping with read/write/execute permissions.
- Interrupts must be disabled (`PSTATE.DAIF=0b1111`).
- The physical memory region occupied by the microhypervisor image must be clean to the PoC.
- All [DMA](#) activity targeting the physical memory region occupied by the microhypervisor must be quiesced. That physical memory region should also be protected against [DMA](#) accesses on systems with an [SMMU](#).

7.1.1.1 Multiboot v2 Launch

Only this launch type supports 64-bit [UEFI](#) platforms.

Register	Value / Description
IP	Physical address of the NOVA ELF image entry point
X0	Multiboot v2 magic value (<code>0x36d76289</code>) [6]
X1	Physical address of the Multiboot v2 information structure [6]
Other	~

The [NOVA](#) microhypervisor consumes the following multiboot tags, if present: 1, 3, 12, 20.

7.1.1.2 Multiboot v1 Launch

Register	Value / Description
IP	Physical address of the NOVA ELF image entry point
X0	Multiboot v1 magic value (<code>0x2badb002</code>) [9]
X1	Physical address of the Multiboot v1 information structure [9]
Other	~

The [NOVA](#) microhypervisor consumes the following multiboot flags, if present: 2, 3.

7.1.1.3 Legacy Launch

Register	Value / Description
IP	Physical address of the NOVA ELF image entry point
X0	Physical address of the Flattened Device Tree (FDT) for the hardware platform [†]
X1	Physical address of the Root Protection Domain (<code>PD_{ROOT}</code>) ELF image
Other	~

[†] Due to its alignment constraint, a valid FDT address will never be equal to a Multiboot magic value.

7.1.2 Root Protection Domain

The [NOVA](#) microhypervisor sets up the [CPU](#) register state as follows when it transfers control to the [Root Execution Context \(EC_{ROOT}\)](#):

Register	Value / Description
IP	HVA of the Root Protection Domain (PD_{ROOT}) ELF image entry point
SP	HVA of the Hypervisor Information Page (HIP)
X0	X0 at boot time [†]
X1	X1 at boot time [†]
X2	X2 at boot time [†]
Other	~

7.2 Memory Map

The [Root Protection Domain \(PD_{ROOT}\)](#) can obtain a list of available/reserved physical memory regions as follows:

- On platforms using Unified Extensible Firmware Interface, by parsing the [UEFI memory map](#).
- On platforms using Flattened Device Tree, by parsing the [FDT](#).

7.3 Class Of Service

Class Of Service ([COS](#)) is currently not supported.

7.4 Protected Resources

Certain resources protected by the [NOVA](#) microhypervisor cannot be delegated and therefore remain inaccessible to user-mode components. The following subsections enumerate these protected resources.

7.4.1 Physical Memory

The following physical memory regions are protected:

- [NOVA](#) microhypervisor – conveyed via [HIP](#).
- [GITS](#), [GICD](#), [GICR](#), [GICC](#), [GICH](#) devices [[10](#), [11](#)] – conveyed via [ACPI](#) MADT or via [FDT](#).
- [SMMU](#) devices [[12](#), [13](#)] – conveyed via [ACPI](#) IORT or via [FDT](#).
- Firmware runtime services – conveyed via [UEFI](#) memory map.

[†]The register contains the preserved original value from the point when control was transferred from the bootloader to the microhypervisor.

7.5 Event-Specific Capability Selectors

For the delivery of exception/intercept messages, the microhypervisor performs an implicit portal traversal.

The selector for the destination portal (**SEL_{OBJ}**):

- is determined by adding the exception/intercept number to the affected **Execution Context**'s Event Selector Base (**SEL_{EVT}**).
- indexes into the **Object Space** (**SPC_{OBJ}**) of the affected **EC**'s **Protection Domain** (**PD**).
- must refer to a **PT Capability** (**CAP_{OBJPT}**) with permission **EVENT** that is bound to an **EC** on the same **CPU** as the affected **EC**, otherwise the affected **EC** is killed.

7.5.1 Architectural Events

Host Exceptions and Guest Intercepts

SEL_{OBJ}	Exception / Intercept	SEL_{OBJ}	Exception / Intercept
SEL_{EVT} + 0x00	Unknown Reason	SEL_{EVT} + 0x20	Instruction Abort (lower EL)
SEL_{EVT} + 0x01	Trapped WFI or WFE	SEL_{EVT} + 0x21	Instruction Abort (same EL)*
SEL_{EVT} + 0x02	reserved	SEL_{EVT} + 0x22	PC Alignment Fault
SEL_{EVT} + 0x03	Trapped MCR or MRC	SEL_{EVT} + 0x23	reserved
SEL_{EVT} + 0x04	Trapped MCRR or MRRC	SEL_{EVT} + 0x24	Data Abort (lower EL)
SEL_{EVT} + 0x05	Trapped MCR or MRC	SEL_{EVT} + 0x25	Data Abort (same EL)*
SEL_{EVT} + 0x06	Trapped LDC or STC	SEL_{EVT} + 0x26	SP Alignment Fault
SEL_{EVT} + 0x07	SME, SVE, SIMD, FPU	SEL_{EVT} + 0x27	Memory Operation Exception
SEL_{EVT} + 0x08	Trapped VMRS Access	SEL_{EVT} + 0x28	Trapped FPU (AArch32)
SEL_{EVT} + 0x09	Trapped PAuth Instruction	SEL_{EVT} + 0x29	reserved
SEL_{EVT} + 0x0a	Trapped LD64B or ST64B	SEL_{EVT} + 0x2a	reserved
SEL_{EVT} + 0x0b	reserved	SEL_{EVT} + 0x2b	reserved
SEL_{EVT} + 0x0c	Trapped MRRC	SEL_{EVT} + 0x2c	Trapped FPU (AArch64)
SEL_{EVT} + 0x0d	Branch Target Exception	SEL_{EVT} + 0x2d	reserved
SEL_{EVT} + 0x0e	Illegal Execution State	SEL_{EVT} + 0x2e	reserved
SEL_{EVT} + 0x0f	reserved	SEL_{EVT} + 0x2f	Error
SEL_{EVT} + 0x10	reserved	SEL_{EVT} + 0x30	Breakpoint (lower EL)
SEL_{EVT} + 0x11	SVC (from AArch32 State)	SEL_{EVT} + 0x31	Breakpoint (same EL)*
SEL_{EVT} + 0x12	HVC (from AArch32 State)	SEL_{EVT} + 0x32	Software Step (lower EL)
SEL_{EVT} + 0x13	SMC (from AArch32 State)	SEL_{EVT} + 0x33	Software Step (same EL)*
SEL_{EVT} + 0x14	reserved	SEL_{EVT} + 0x34	Watchpoint (lower EL)
SEL_{EVT} + 0x15	SVC (from AArch64 State)*	SEL_{EVT} + 0x35	Watchpoint (same EL)*
SEL_{EVT} + 0x16	HVC (from AArch64 State)	SEL_{EVT} + 0x36	reserved
SEL_{EVT} + 0x17	SMC (from AArch64 State)	SEL_{EVT} + 0x37	reserved
SEL_{EVT} + 0x18	Trapped MSR or MRS	SEL_{EVT} + 0x38	BKPT (AArch32)
SEL_{EVT} + 0x19	Trapped SVE	SEL_{EVT} + 0x39	reserved
SEL_{EVT} + 0x1a	Trapped ERET	SEL_{EVT} + 0x3a	Vector Catch (AArch32)
SEL_{EVT} + 0x1b	TSTART Exception	SEL_{EVT} + 0x3b	reserved
SEL_{EVT} + 0x1c	PAuth Instruction Failure	SEL_{EVT} + 0x3c	BRK (AArch64)
SEL_{EVT} + 0x1d	Trapped SME	SEL_{EVT} + 0x3d	reserved
SEL_{EVT} + 0x1e	Granule Protection Exception	SEL_{EVT} + 0x3e	reserved
SEL_{EVT} + 0x1f	reserved	SEL_{EVT} + 0x3f	reserved

Please refer to [3] for more details on each of these events.

*These events may be handled by the microhypervisor, in which case they will not cause portal traversals.

7.5.2 Microhypervisor Events

SEL_{OBJ}	Event
SEL_{EVT} + SEL _{ARCH} + 0x0	Startup
SEL_{EVT} + SEL _{ARCH} + 0x1	Recall
SEL_{EVT} + SEL _{ARCH} + 0x2	Virtual Timer

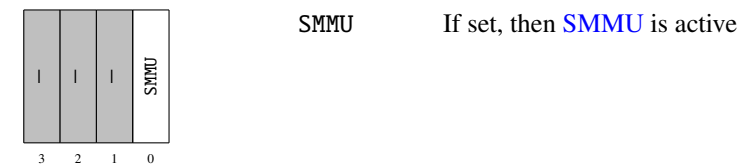
The value of SEL_{ARCH} depends on the origin of the event:

- SEL_{ARCH} = SEL_{HST/ARCH} for events that occurred in the host.
- SEL_{ARCH} = SEL_{GST/ARCH} for events that occurred in the guest.

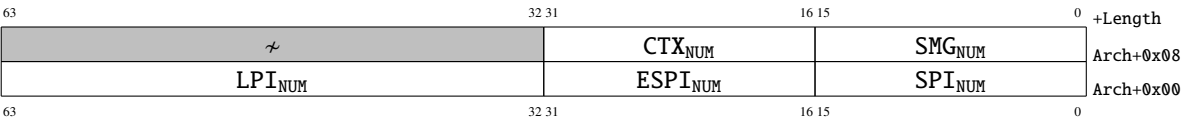
7.6 Architecture-Dependent Structures

7.6.1 Platform Features

The supported platform features are defined as follows:



7.6.2 Hypervisor Information Page



- SPI_{NUM}

Total number of Shared Peripheral Interrupts (SPIs).
- ESPI_{NUM}

Total number of Extended Shared Peripheral Interrupts (ESPIs).
- LPI_{NUM}

Total number of Locality-Specific Peripheral Interrupts (LPIs).
- SMG_{NUM}

Total number of SMMUv2 Stream Mapping Groups (SMGs).
- CTX_{NUM}

Total number of SMMUv2 Translation Contexts (CTXs).

7.6.3 User Thread Control Block

-		SEL_GST		+0x2e0	
-		VMCR	ELRSR	+0x2d0	
AP1R3	AP1R2	AP1R1	AP1R0	+0x2c0	
AP0R3	AP0R2	AP0R1	AP0R0	+0x2b0	
LR15		LR14		+0x2a0	
LR13		LR12		+0x290	
LR11		LR10		+0x280	
LR9		LR8		+0x270	
LR7		LR6		+0x260	
LR5		LR4		+0x250	
LR3		LR2		+0x240	
LR1		LR0		+0x230	
CNTVOFF_EL2		CNTKCTL_EL1		+0x220	
CNTV_CTL_EL0		CNTV_CVAL_EL0		+0x210	
-		HPFAR_EL2		+0x200	
FAR_EL2		ESR_EL2		+0x1f0	
SPSR_EL2		ELR_EL2		+0x1e0	
VMPIDR_EL2		VPIDR_EL2		+0x1d0	
HCRX_EL2		HCR_EL2		+0x1c0	
-		MDSCR_EL1		+0x1b0	
SCTLR_EL1		VBAR_EL1		+0x1a0	
AMAIR_EL1		MAIR_EL1		+0x190	
TCR_EL1		TTBR1_EL1		+0x180	
TTBR0_EL1		AFSR1_EL1		+0x170	
AFSR0_EL1		FAR_EL1		+0x160	
ESR_EL1		SPSR_EL1		+0x150	
ELR_EL1		CONTEXTIDR_EL1		+0x140	
TPIDR_EL1		SP_EL1		+0x130	
-	HSTR	IFSR	DACR	+0x120	
SPSR_und	SPSR_irq	SPSR_fiq	SPSR_abt	+0x110	
TPIDRRO_EL0		TPIDR_EL0		+0x100	
SP_EL0		X30 (LR_fiq)		+0x0f0	
X29 (SP_fiq)		X28 (R12_fiq)		+0x0e0	
X27 (R11_fiq)		X26 (R10_fiq)		+0x0d0	
X25 (R9_fiq)		X24 (R8_fiq)		+0x0c0	
X23 (SP_und)		X22 (LR_und)		+0x0b0	
X21 (SP_abt)		X20 (LR_abt)		+0x0a0	
X19 (SP_svc)		X18 (LR_svc)		+0x090	
X17 (SP_irq)		X16 (LR_irq)		+0x080	
X15 (SP_hyp)		X14 (LR_usr)		+0x070	
X13 (SP_usr)		X12 (R12_usr)		+0x060	
X11 (R11_usr)		X10 (R10_usr)		+0x050	
X9 (R9_usr)		X8 (R8_usr)		+0x040	
X7 (R7)		X6 (R6)		+0x030	
X5 (R5)		X4 (R4)		+0x020	
X3 (R3)		X2 (R2)		+0x010	
X1 (R1)		X0 (R0)		+0x000	

7.6.4 Message Transfer Descriptor

The [Message Transfer Descriptor \(MTD\)](#), which controls the subset of the architectural state transferred during exceptions and intercepts, as described in Section 4.4.2, has the following layout:

SPACES	GIC	TMR	—	EL2_HPFAR	EL2_ESR_FAR	EL2_ELR_SPSR	EL2_IDR	EL2_HCR	—	EL1_MDSCR	EL1_SCTLR	EL1_VBAR	EL1_MAIR	EL1_TCR	EL1_TTBR	EL1_AFSR	EL1_ESR_FAR	EL1_ELR_SPSR	EL1_IDR	EL1_SP	—	A32_DIH	A32_SPSR	—	EL0_IDR	EL0_SP	FPR	GPR	ICI	POISON
31	30	29		27	26	25	24	23		20	19	18	17	16	15	14	13	12	11	10		8	7		5	4	3	2	1	0

Each [MTD](#) bit controls the transfer of the listed architectural state to/from the respective fields in the [UTCB](#) (7.6.3) as follows:

- State with access **r** can be read from the architectural state into the [UTCB](#).
- State with access **w** can be written from the [UTCB](#) into the architectural state.

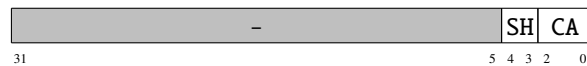
MTD Bit	Access	Host Execution Context State	Guest Execution Context State
POISON	w	Kills the EC_{HST}	Kills the EC_{GST}
ICI [†]	w	Invalidates the entire I-Cache	Invalidates the entire I-Cache
GPR	rw	X0 ... X30	X0 ... X30
EL0_SP	rw	SP_EL0	SP_EL0
EL0_IDR	rw	TPIDR_EL0, TPIDRRO_EL0	TPIDR_EL0, TPIDRRO_EL0
A32_SPSR	rw	—	SPSR_ABT, SPSR_FIQ, SPSR_IRQ, SPSR_UND
A32_DIH	rw	—	DACR, IFSR, HSTR
EL1_SP	rw	—	SP_EL1
EL1_IDR	rw	—	TPIDR_EL1, CONTEXTIDR_EL1
EL1_ELR_SPSR	rw	—	ELR_EL1, SPSR_EL1
EL1_ESR_FAR	rw	—	ESR_EL1, FAR_EL1
EL1_AFSR	rw	—	AFSR0_EL1, AFSR1_EL1
EL1_TTBR	rw	—	TTBR0_EL1, TTBR1_EL1
EL1_TCR	rw	—	TCR_EL1
EL1_MAIR	rw	—	MAIR_EL1, AMAIR_EL1
EL1_VBAR	rw	—	VBAR_EL1
EL1_SCTLR	rw	—	SCTLR_EL1
EL1_MDSCR	rw	—	MDSCR_EL1
EL2_HCR	rw	—	HCR_EL2, HCRX_EL2
EL2_IDR	rw	—	VPIDR_EL2, VMPIDR_EL2
EL2_ELR_SPSR	rw	ELR_EL2, SPSR_EL2*	ELR_EL2, SPSR_EL2
EL2_ESR_FAR	r	ESR_EL2, FAR_EL2	ESR_EL2, FAR_EL2
EL2_HPFAR	r	—	HPFAR_EL2
TMR	rw	—	CNTV_CVAL_EL0, CNTV_CTL_EL0 CNTKCTL_EL1, CNTVOFF_EL2
GIC	rw r	—	LR0 ... LR15, APxR0 ... APxR3 ELRSR, VMCR
SPACES	w	—	SEL_GST

*Only the condition flags are writable.

[†]Only affects a VIPT instruction cache of the local [CPU](#). Has no effect on PIPT instruction caches, data caches, or caches of other [CPUs](#).

7.6.5 Memory Attribute Descriptor

The Memory Attribute Descriptor (**MAD**) describes page attributes for a **CAP_{MEM}** delegation from **SPC_{HSTNOVA}**.



The fields are defined as follows:

CA

Specifies the cacheability attributes of the page according to the following table:

Encoding	Cacheability	Description
0x0	DEV	Device
0x1	DEV_E	Device, Early Ack
0x2	DEV_RE	Device, Early Ack, Reordering
0x3	DEV_GRE	Device, Early Ack, Reordering, Gathering
0x4	-	<i>reserved</i>
0x5	MEM_NC	Memory, Inner/Outer Non-Cacheable
0x6	MEM_WT	Memory, Inner/Outer Write-Through
0x7	MEM_WB	Memory, Inner/Outer Write-Back

SH

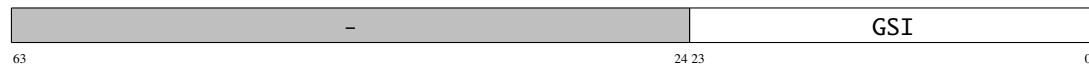
Specifies the shareability attributes of the page according to the following table:

Encoding	Shareability	Description
0x0	NONE	Not Shareable
0x1	-	<i>reserved</i>
0x2	OUTER	Outer Shareable
0x3	INNER	Inner Shareable

Please refer to [3] for the architectural behavior of these attributes.

7.6.6 Interrupt Semaphore Descriptor

The Interrupt Semaphore Descriptor (**ISD**) describes the interrupt associated with an Interrupt **Semaphore**.



The fields are defined as follows:

GSI

Designates the interrupt via its **GSI** number (INTID) in one of the following ranges:

- [32, 32+**SPI_{NUM}**) – Shared Peripheral Interrupt (**SPI**)
- [4096, 4096+**ESPI_{NUM}**) – Extended Shared Peripheral Interrupt (**ESPI**)
- [8192, 8192+**LPI_{NUM}**) – Locality-Specific Peripheral Interrupt (**LPI**)

7.6.7 Device Topology Descriptor

The Device Topology Descriptor (DTD) describes the topology of a device as follows:

SID

Stream Identifier (SID) – if the device is subject to DMA translation.

SID Mask

Specifies which bits of that SID should be matched (0) or ignored (1) by the Stream Mapping Group.

SMG

Specifies the SMMU Stream Mapping Group (SMG) to use for that SID – must be < SMG_{NUM}.

CTX

Specifies the SMMU Translation Context (CTX) to use for that SMG – must be < CTX_{NUM}.

DID

Device Identifier (DID) – if the device is subject to MSI translation.

SMMU

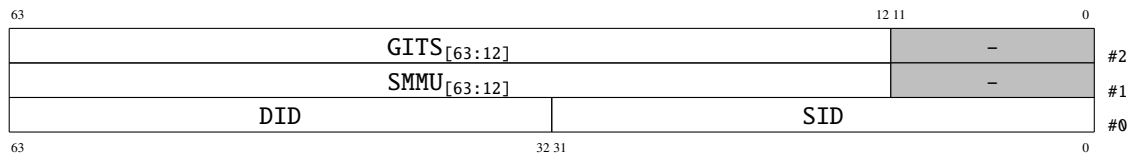
Host-Physical Address (HPA) of the SMMU that performs DMA translation for the device.

GITS

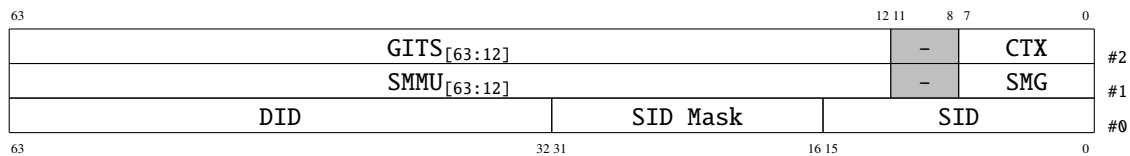
Host-Physical Address (HPA) of the GITS that performs MSI translation for the device.

Reserved fields and fields that do not apply to a particular platform must be zero.

7.6.7.1 SMMUv3



7.6.7.2 SMMUv2



System software must ensure an unambiguous assignment of Stream Identifiers to Stream Mapping Groups, i.e. it must configure the SID/Mask fields across all Stream Mapping Groups such that no SID multi-matches can occur.

7.7 Calling Convention

The following pages describes the calling convention for each hypercall. A [Host Execution Context](#) calls into the microhypervisor by loading the hypercall identifier and other parameters into the specified [CPU](#) registers and then executes the `svc #0` instruction [3].

The hypercall identifier consists of the [hypercall number](#) and hypercall-specific flags, as illustrated in Figure 7.1.

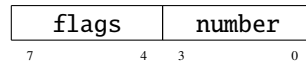


Figure 7.1: Hypercall Identifier

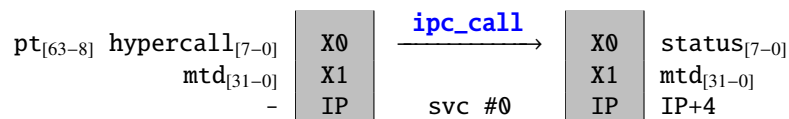
The [status code](#) returned from a hypercall has the format shown in Figure 7.2.



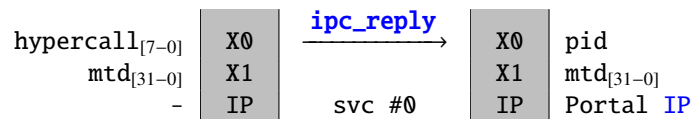
Figure 7.2: Status Code

The assignment of hypercall parameters to [CPU](#) registers is shown on the left side, the contents of the [CPU](#) registers after the hypercall is shown on the right side.

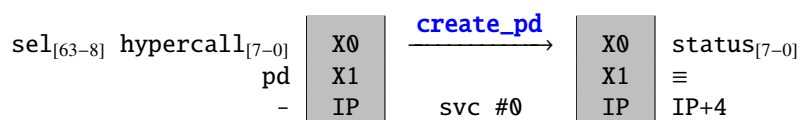
IPC Call



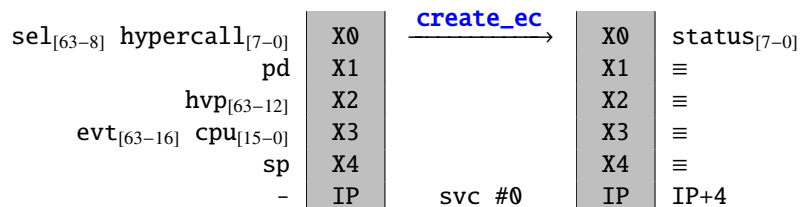
IPC Reply



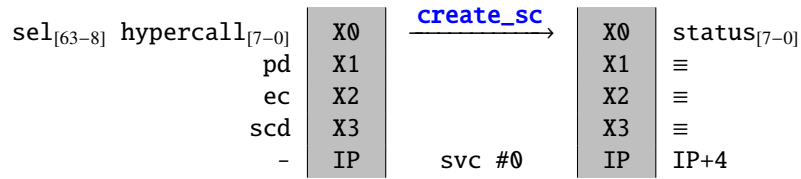
Create Protection Domain



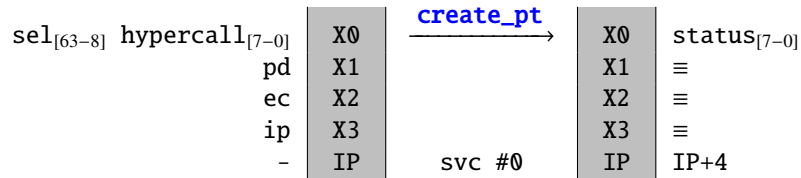
Create Execution Context



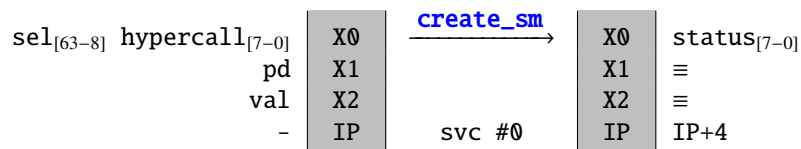
Create Scheduling Context



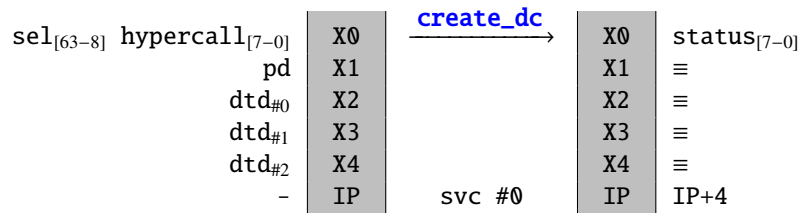
Create Portal



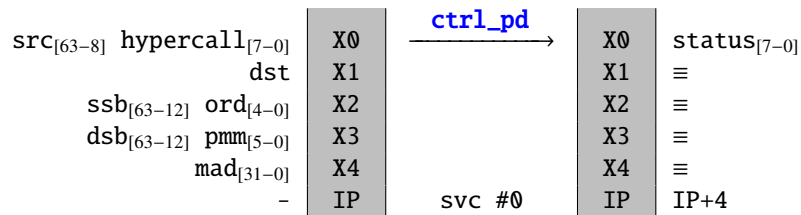
Create Semaphore



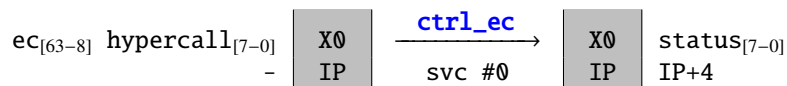
Create Device Context



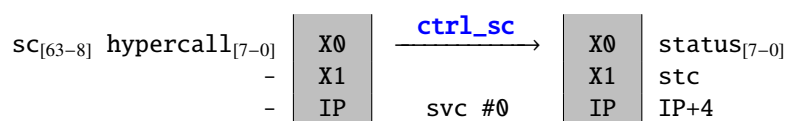
Control Protection Domain



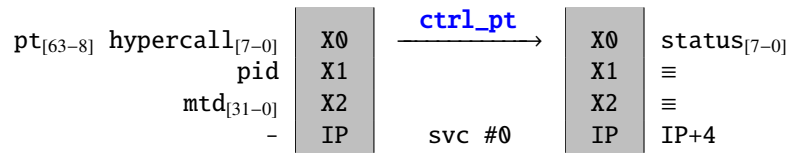
Control Execution Context



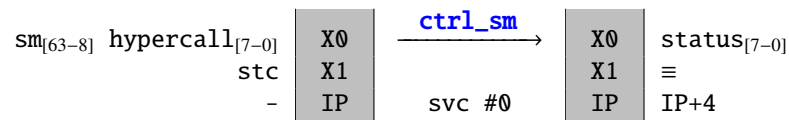
Control Scheduling Context



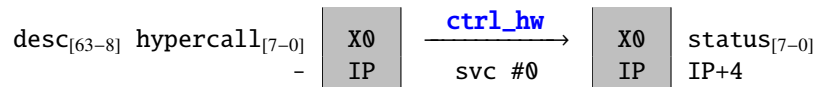
Control Portal



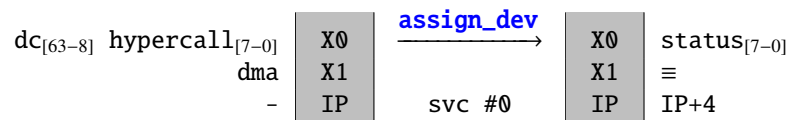
Control Semaphore



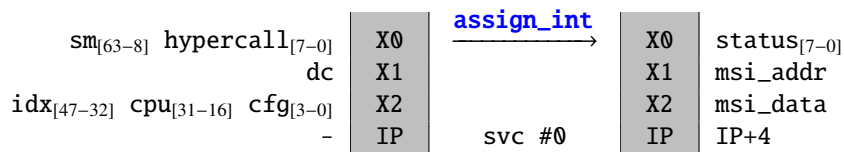
Control Hardware



Assign Device



Assign Interrupt



7.8 Supplementary Functionality

This section describes functions that do **not** conform to the calling convention for hypercalls. Because these functions cannot perform capability-based access control, their invocation is restricted to the [Root Protection Domain](#) (PD_{ROOT}). Invocation of these functions from any other [Protection Domain](#) generates an exception.

Secure Monitor Call

identifier _[31-0]		proxy_smc →		
-	X0		X0	~
-	X1		X1	~
-	X2		X2	~
-	X3		X3	~
-	X4		X4	~
-	X5		X5	~
-	X6		X6	~
-	X7		X7	~
-	X8		X8	~
-	X9		X9	~
-	X10		X10	~
-	X11		X11	~
-	X12		X12	~
-	X13		X13	~
-	X14		X14	~
-	X15		X15	~
-	X16		X16	~
-	X17		X17	~
-	IP	svc #1	IP	IP+4

This call is proxy-filtered by the microhypervisor. If the combination of invoked service (identifier_[29-24]) and function (identifier_[15-0]) is listed in the table below, then the microhypervisor issues the corresponding SMC to platform firmware on behalf of the caller. Otherwise, this function generates an exception. Register allocation conforms to the Arm SMCCC [14].

Service	Description	Function	Description
0x2	SIP Service Calls	0x0000-0xffff	All functions
0x4	Standard Secure Service Calls	0x0050-0x005f	TRNG functions [15]
		0x0130-0x013f	PCI functions [16]
0x30-0x31	Trusted Application Calls	0x0000-0xffff	All functions
0x32-0x3f	Trusted OS Calls	0x0000-0xffff	All functions

8 ABI x86-64

8.1 Boot State

8.1.1 NOVA Microhypervisor

The bootloader must set up the [CPU](#) register state according to one of the launch types listed below when it transfers control to the [NOVA](#) microhypervisor entry point. Furthermore, the following preconditions must be satisfied:

- The [CPU](#) state must conform to a machine state defined in the Multiboot Specification v2 [6] or v1 [9].
- All [DMA](#) activity targeting the physical memory region occupied by the microhypervisor must be quiesced. That physical memory region should also be protected against [DMA](#) accesses on systems with an [IOMMU](#).

8.1.1.1 Multiboot v2 Launch

Only this launch type supports 64-bit [UEFI](#) platforms.

Register	Value / Description
EIP	Physical address of the NOVA ELF image entry point
EAX	Multiboot v2 magic value (0x36d76289) [6]
EBX	Physical address of the Multiboot v2 information structure [6]
Other	↯

The [NOVA](#) microhypervisor consumes the following multiboot tags, if present: 1, 3, 12, 20.

8.1.1.2 Multiboot v1 Launch

Register	Value / Description
EIP	Physical address of the NOVA ELF image entry point
EAX	Multiboot v1 magic value (0x2badb002) [9]
EBX	Physical address of the Multiboot v1 information structure [9]
Other	↯

The [NOVA](#) microhypervisor consumes the following multiboot flags, if present: 2, 3.

8.1.2 Root Protection Domain

The [NOVA](#) microhypervisor sets up the [CPU](#) register state as follows when it transfers control to the [Root Execution Context](#) (EC_{ROOT}):

Register	Value / Description
RIP	HVA of the Root Protection Domain (PD_{ROOT}) ELF image entry point
RSP	HVA of the Hypervisor Information Page (HIP)
RDI	EAX at boot time [†]
RSI	EBX at boot time [†]
Other	↯

[†]The register contains the preserved original value from the point when control was transferred from the bootloader to the microhypervisor.

8.2 Memory Map

The [Root Protection Domain](#) (PD_{ROOT}) can obtain a list of available/reserved physical memory regions as follows:

- On platforms using Multiboot v2 (UEFI boot services enabled), by parsing the [UEFI memory map](#) [8].
- On platforms using Multiboot v2, by parsing the [Multiboot v2](#) memory map [6].
- On platforms using Multiboot v1, by parsing the [Multiboot v1](#) memory map [9].

8.3 Class Of Service

Class Of Service ([COS](#)) support is indicated by CPUID leaf $0x7$, sub-leaf $0x0$: $EBX_{[15]}$.

The [Root Protection Domain](#) (PD_{ROOT}) must perform the following steps on each [CPU](#) to configure [QOS](#) settings:

1. Invoke [ctrl_hw](#) to establish a valid [QOS](#) configuration:
 - CDP_{L3} support is indicated by CPUID leaf $0x10$, sub-leaf $0x1$: $ECX_{[2]}$
 - CDP_{L2} support is indicated by CPUID leaf $0x10$, sub-leaf $0x2$: $ECX_{[2]}$
2. Determine COS_{NUM} as the maximum of the following:
 - COS_{L3} from CPUID leaf $0x10$, sub-leaf $0x1$: $(1+EDX_{[15-0]}) >> X$, where
 - $X=0$ if CDP_{L3} is disabled.
 - $X=1$ if CDP_{L3} is enabled.
 - COS_{L2} from CPUID leaf $0x10$, sub-leaf $0x2$: $(1+EDX_{[15-0]}) >> X$, where
 - $X=0$ if CDP_{L2} is disabled.
 - $X=1$ if CDP_{L2} is enabled.
 - COS_{MB} from CPUID leaf $0x10$, sub-leaf $0x3$: $(1+EDX_{[15-0]})$
3. Invoke [ctrl_hw](#) to configure the following:
 - For each COS_{L3} : [CAT/CDP](#) L3 Capacity Bitmask(s)
 - For each COS_{L2} : [CAT/CDP](#) L2 Capacity Bitmask(s)
 - For each COS_{MB} : [MBA](#) Delay

8.4 Protected Resources

Certain resources protected by the [NOVA](#) microhypervisor cannot be delegated and therefore remain inaccessible to user-mode components. The following subsections enumerate these protected resources.

8.4.1 Physical Memory

The following physical memory regions are protected:

- [NOVA](#) microhypervisor – conveyed via [HIP](#).
- [LAPIC](#), [IOAPIC](#) devices – conveyed via [ACPI](#) MADT.
- [IOMMU](#) devices [17, 18] – conveyed via [ACPI](#) DMAR or IVRS.
- [DPR](#) and [TXT](#) memory [19], except for [TPM](#) localities 0 and 1.
- [MSI](#) range – at $0xfeef000000$
- Firmware runtime services – conveyed via [UEFI](#) memory map.

8.4.2 I/O Ports

The **CAP_{PIO}** column in the table below details the permissions granted for I/O ports by the **NOVA PIO Space**.

I/O Port	CAP _{PIO}	Exception or I/O Exit	Enumeration	VMM Handling
PM1a_CNT PM1b_CNT PM2_CNT SMI_CMD	CAP ₀	Always	ACPI FADT	Emulate
Other	A	If A is not set	–	Emulate or Passthrough

8.4.3 Model-Specific Registers

The **CAP_{MSR}** column in the table below details the permissions granted for **MSRs** by the **NOVA MSR Space**.

Model-Specific Register	CAP _{MSR}	RDMSR Exit	WRMSR Exit	VMM Handling
IA32_SYSENTER_CS* IA32_SYSENTER_ESP* IA32_SYSENTER_EIP* IA32_PAT* IA32_EFER* IA32_FS_BASE* IA32_GS_BASE* IA32_KERNEL_GS_BASE*	RW	If R is not set	If W is not set	Emulate or Passthrough
IA32_PRED_CMD IA32_FLUSH_CMD	W	Always	If W is not set	
IA32_TSC IA32_PLATFORM_ID IA32_BIOS_SIGN_ID IA32_SGXLEPUBKEYHASH{0-3}* IA32_CORE_CAPABILITIES IA32_MPERF IA32_APERF IA32_MTRR_CAP IA32_ARCH_CAPABILITIES IA32_MCU_OPT_CTRL IA32_OVERCLOCKING_STATUS IA32_PERF_STATUS IA32_THERM_STATUS IA32_PACKAGE_THERM_STATUS IA32_SGX_SVN_STATUS IA32_XSS* IA32_STAR* IA32_LSTAR* IA32_FMASK* IA32_TSC_AUX*	R	If R is not set	Always	
Other	CAP ₀	Always	Always	Emulate

*The VMM can read/write the guest-effective **MSR** value via the **UTCB**.

8.5 Event-Specific Capability Selectors

For the delivery of exception/intercept messages, the microhypervisor performs an implicit portal traversal.

The selector for the destination portal (SEL_{OBJ}):

- is determined by adding the exception/intercept number to the affected [Execution Context](#)'s Event Selector Base (SEL_{EVT}).
- indexes into the [Object Space](#) (SPC_{OBJ}) of the affected [EC](#)'s [Protection Domain](#) ([PD](#)).
- must refer to a [PT Capability](#) (CAP_{OBJPT}) with permission `EVENT` that is bound to an [EC](#) on the same [CPU](#) as the affected [EC](#), otherwise the affected [EC](#) is killed.

8.5.1 Architectural Events

Host Exceptions

SEL_{OBJ}	Exception	SEL_{OBJ}	Exception
$SEL_{EVT} + 0x00$	#DE	$SEL_{EVT} + 0x10$	#MF
$SEL_{EVT} + 0x01$	#DB	$SEL_{EVT} + 0x11$	#AC
$SEL_{EVT} + 0x02$	reserved	$SEL_{EVT} + 0x12$	#MC*
$SEL_{EVT} + 0x03$	#BP	$SEL_{EVT} + 0x13$	#XM
$SEL_{EVT} + 0x04$	#OF	$SEL_{EVT} + 0x14$	#VE
$SEL_{EVT} + 0x05$	#BR	$SEL_{EVT} + 0x15$	#CP
$SEL_{EVT} + 0x06$	#UD	$SEL_{EVT} + 0x16$	reserved
$SEL_{EVT} + 0x07$	#NM*	$SEL_{EVT} + 0x17$	reserved
$SEL_{EVT} + 0x08$	#DF*	$SEL_{EVT} + 0x18$	reserved
$SEL_{EVT} + 0x09$	reserved	$SEL_{EVT} + 0x19$	reserved
$SEL_{EVT} + 0x0a$	#TS*	$SEL_{EVT} + 0x1a$	reserved
$SEL_{EVT} + 0x0b$	#NP	$SEL_{EVT} + 0x1b$	reserved
$SEL_{EVT} + 0x0c$	#SS	$SEL_{EVT} + 0x1c$	reserved
$SEL_{EVT} + 0x0d$	#GP	$SEL_{EVT} + 0x1d$	reserved
$SEL_{EVT} + 0x0e$	#PF	$SEL_{EVT} + 0x1e$	reserved
$SEL_{EVT} + 0x0f$	reserved	$SEL_{EVT} + 0x1f$	reserved

*These events may be handled by the microhypervisor, in which case they will not cause portal traversals.

†These events may be force-enabled by the microhypervisor, in which case they will cause portal traversals.

Guest Intercepts (VMX)

SEL_{OBJ}	Intercept	SEL_{OBJ}	Intercept
SEL_{EVT} + 0x00	Exception or NMI*	SEL_{EVT} + 0x30	EPT Violation†
SEL_{EVT} + 0x01	External Interrupt*	SEL_{EVT} + 0x31	EPT Misconfiguration
SEL_{EVT} + 0x02	Triple Fault†	SEL_{EVT} + 0x32	INVEPT
SEL_{EVT} + 0x03	INIT†	SEL_{EVT} + 0x33	RDTSCP
SEL_{EVT} + 0x04	SIPI†	SEL_{EVT} + 0x34	Preemption Timer
SEL_{EVT} + 0x05	I/O SMI	SEL_{EVT} + 0x35	INVVPID
SEL_{EVT} + 0x06	Other SMI	SEL_{EVT} + 0x36	WBINVD, WBNOINVD
SEL_{EVT} + 0x07	Interrupt Window	SEL_{EVT} + 0x37	XSETBV
SEL_{EVT} + 0x08	NMI Window	SEL_{EVT} + 0x38	APIC Write
SEL_{EVT} + 0x09	Task Switch†	SEL_{EVT} + 0x39	RDRAND
SEL_{EVT} + 0x0a	CPUID†	SEL_{EVT} + 0x3a	INVPCID
SEL_{EVT} + 0x0b	GETSEC†	SEL_{EVT} + 0x3b	VMFUNC
SEL_{EVT} + 0x0c	HLT†	SEL_{EVT} + 0x3c	ENCLS
SEL_{EVT} + 0x0d	INVD†	SEL_{EVT} + 0x3d	RDSEED
SEL_{EVT} + 0x0e	INVLPG	SEL_{EVT} + 0x3e	PML Log Full
SEL_{EVT} + 0x0f	RDPMC	SEL_{EVT} + 0x3f	XSAVES
SEL_{EVT} + 0x10	RDTSC	SEL_{EVT} + 0x40	XRSTORS
SEL_{EVT} + 0x11	RSM	SEL_{EVT} + 0x41	PCONFIG
SEL_{EVT} + 0x12	VMCALL	SEL_{EVT} + 0x42	SPP Miss / Misconfiguration
SEL_{EVT} + 0x13	VMCLEAR	SEL_{EVT} + 0x43	UMWAIT
SEL_{EVT} + 0x14	VMLAUNCH	SEL_{EVT} + 0x44	TPAUSE
SEL_{EVT} + 0x15	VMPTRLD	SEL_{EVT} + 0x45	LOADIWKEY
SEL_{EVT} + 0x16	VMPTRST	SEL_{EVT} + 0x46	ENCLV
SEL_{EVT} + 0x17	VMREAD	SEL_{EVT} + 0x47	SGX Conflict
SEL_{EVT} + 0x18	VMRESUME	SEL_{EVT} + 0x48	ENQCMD PASID Failure
SEL_{EVT} + 0x19	VMWRITE	SEL_{EVT} + 0x49	ENQCMDS PASID Failure
SEL_{EVT} + 0x1a	VMXOFF	SEL_{EVT} + 0x4a	Bus Lock
SEL_{EVT} + 0x1b	VMXON	SEL_{EVT} + 0x4b	Instruction Timeout
SEL_{EVT} + 0x1c	CR Access*	SEL_{EVT} + 0x4c	SEAMCALL
SEL_{EVT} + 0x1d	DR Access	SEL_{EVT} + 0x4d	TDCALL
SEL_{EVT} + 0x1e	I/O Access†	SEL_{EVT} + 0x4e	RDMSRLIST
SEL_{EVT} + 0x1f	RDMSR†	SEL_{EVT} + 0x4f	WRMSRLIST
SEL_{EVT} + 0x20	WRMSR†	SEL_{EVT} + 0x50	URDMSR
SEL_{EVT} + 0x21	VM Entry Failure (State)†	SEL_{EVT} + 0x51	UWRMSR
SEL_{EVT} + 0x22	VM Entry Failure (MSR)	SEL_{EVT} + 0x52	reserved
SEL_{EVT} + 0x23	reserved	SEL_{EVT} + 0x53	reserved
SEL_{EVT} + 0x24	MWAIT	SEL_{EVT} + 0x54	reserved
SEL_{EVT} + 0x25	MTF	SEL_{EVT} + 0x55	reserved
SEL_{EVT} + 0x26	reserved	SEL_{EVT} + 0x56	reserved
SEL_{EVT} + 0x27	MONITOR	SEL_{EVT} + 0x57	reserved
SEL_{EVT} + 0x28	PAUSE	SEL_{EVT} + 0x58	reserved
SEL_{EVT} + 0x29	VM Entry Failure (MCE)	SEL_{EVT} + 0x59	reserved
SEL_{EVT} + 0x2a	reserved	SEL_{EVT} + 0x5a	reserved
SEL_{EVT} + 0x2b	TPR Below Threshold	SEL_{EVT} + 0x5b	reserved
SEL_{EVT} + 0x2c	APIC Access	SEL_{EVT} + 0x5c	reserved
SEL_{EVT} + 0x2d	Virtualized EOI	SEL_{EVT} + 0x5d	reserved
SEL_{EVT} + 0x2e	GDTR/IDTR Access	SEL_{EVT} + 0x5e	reserved
SEL_{EVT} + 0x2f	LDTR/TR Access	SEL_{EVT} + 0x5f	reserved

Please refer to [4] for more details on each of these events.

8.5.2 Microhypervisor Events

SEL_{OBJ}	Event
SEL_{EVT} + SEL _{ARCH} + 0x0	Startup
SEL_{EVT} + SEL _{ARCH} + 0x1	Recall

The value of SEL_{ARCH} depends on the origin of the event:

- SEL_{ARCH} = SEL_{HST/ARCH} for events that occurred in the host.
- SEL_{ARCH} = SEL_{GST/ARCH} for events that occurred in the guest.

8.6 Architecture-Dependent Structures

8.6.1 Platform Features

The supported platform features are defined as follows:

I	SVM	VMX	IOMMU	IOMMU	If set, then IOMMU is active
				VMX	If set, then Intel VMX is active
				SVM	If set, then AMD SVM is active
3	2	1	0		

8.6.2 Hypervisor Information Page

63	32 31				0				+Length	
Event Log Offset					Event Log Size					Arch+0x10
Event Log Physical Address										Arch+0x08
GSI _{NUM}					PIN _{NUM}				VEC _{NUM}	Arch+0x00
63	32 31				16 15		8 7		0	

VEC_{NUM}

Total number of per-CPU interrupt vectors.

PIN_{NUM}

Total number of interrupts that are type PIN in [GSI](#) range [0, PIN_{NUM}). All other interrupts are type [MSI](#).

GSI_{NUM}

Total number of Global System Interrupts ([GSIs](#)).

Event Log Physical Address

Page-aligned physical address of the [TPM](#) event log [20].

Event Log Size

Size of the [TPM](#) event log in bytes.

Event Log Offset

Offset of the byte after the last [TPM](#) event log entry.

8.6.3 User Thread Control Block

SEL_MSR		SEL_PIO		+0x280	
SEL_GST		IA32_TSC_AUX		+0x270	
IA32_KERNEL_GS_BASE		IA32_FMASK		+0x260	
IA32_LSTAR		IA32_STAR		+0x250	
IA32_EFER		IA32_PAT		+0x240	
IA32_SYSENTER_EIP		IA32_SYSENTER_ESP		+0x230	
IA32_SYSENTER_CS		IA32_APIC_BASE		+0x220	
IA32_SGXLEPUBKEYHASH3		IA32_SGXLEPUBKEYHASH2		+0x210	
IA32_SGXLEPUBKEYHASH1		IA32_SGXLEPUBKEYHASH0		+0x200	
IA32_XSS		XCR0		+0x1f0	
DR7		CR8		+0x1e0	
CR4		CR3		+0x1d0	
CR2		CR0		+0x1c0	
PDPTE3		PDPTE2		+0x1b0	
PDPTE1		PDPTE0		+0x1a0	
Base IDTR		Limit IDTR	-		+0x190
Base GDTR		Limit GDTR	-		+0x180
Base LDTR		Limit LDTR	AR LDTR*	SEL LDTR	+0x170
Base TR		Limit TR	AR TR*	SEL TR	+0x160
Base GS		Limit GS	AR GS*	SEL GS	+0x150
Base FS		Limit FS	AR FS*	SEL FS	+0x140
Base ES		Limit ES	AR ES*	SEL ES	+0x130
Base DS		Limit DS	AR DS*	SEL DS	+0x120
Base SS		Limit SS	AR SS*	SEL SS	+0x110
Base CS		Limit CS	AR CS*	SEL CS	+0x100
IDT Vectoring Error	IDT Vectoring Info	Interruption Error	Interruption Info [†]		+0x0f0
TPR Threshold	PF Error Match	PF Error Mask	EXC Intercepts		+0x0e0
CR4 Intercepts		CR0 Intercepts			+0x0d0
3rd Exec Controls		2nd Exec Controls	1st Exec Controls		+0x0c0
-		3rd Qualification			+0x0b0
2nd Qualification		1st Qualification			+0x0a0
Activity	Interruptibility	Instruction Info	Instruction Length		+0x090
RIP		RFLAGS			+0x080
R15		R14			+0x070
R13		R12			+0x060
R11		R10			+0x050
R9		R8			+0x040
R7 (RDI)		R6 (RSI)			+0x030
R5 (RBP)		R4 (RSP)			+0x020
R3 (RBX)		R2 (RDX)			+0x010
R1 (RCX)		R0 (RAX)			+0x000

*See Section 8.6.3.1 for encoding details.

†See Section 8.6.3.2 for encoding details.

8.6.3.1 Encoding: Segment Access Rights

–	U	G	D/B	L	AVL	P	DPL	S	Type
	12	11	10	9	8	7	6	5	4
							3		0
Field	Description								
U	0 = Segment Usable 1 = Segment Unusable								
G	Granularity								
D/B	0 = 16-bit segment 1 = 32-bit segment								
L	64-bit mode active (CS only)								
AVL	Available for use by system software								
P	Segment Present								
DPL	Descriptor Privilege Level								
S	0 = System 1 = Code or Data								
Type	Segment Type								

8.6.3.2 Encoding: Interruption Information

V											N	I	E	Type	Vector						
31											13 12 11 10 8 7 0										
Field											Description										
V											0 = Fields E, Type, Vector are invalid 1 = Fields E, Type, Vector are valid										
N											0 = Do not request an NMI window 1 = Request an NMI window										
I											0 = Do not request an interrupt window 1 = Request an interrupt window										
E											0 = Do not deliver the error code from the UTCB Interruption Error field 1 = Deliver the error code from the UTCB Interruption Error field										
Type											0 = External Interrupt 2 = Non-Maskable Interrupt 3 = Hardware Exception 4 = Software Interrupt 5 = Privileged Software Exception 6 = Software Exception 7 = Other Event (not delivered through IDT)										
Vector											IDT Vector of Interrupt or Exception										

8.6.4 Message Transfer Descriptor

The [Message Transfer Descriptor \(MTD\)](#), which controls the subset of the architectural state transferred during exceptions and intercepts, as described in Section 4.4.2, has the following layout:

SPACES	FPU	TLB	TSC	KERNEL_GS	EFER	PAT	SYSENTER	SYSCALL	APIC	SGX	XSAVE	DR	CR	PDPTE	IDTR	GDTR	LDTR	TR	FS/GS	DS/ES	CS/SS	INJ	TPR	CTRL	QUAL	STA	RIP	RFLAGS	GPR ₈₋₁₅	GPR ₀₋₇	POISON
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Each [MTD](#) bit controls the transfer of the listed architectural state to/from the respective fields in the [UTCB](#) (8.6.3) as follows:

- State with access **r** can be read from the architectural state into the [UTCB](#).
- State with access **w** can be written from the [UTCB](#) into the architectural state.

MTD Bit	Access	Host Execution Context State	Guest Execution Context State
POISON	w	Kills the EC_{HST}	Kills the EC_{GST}
GPR ₀₋₇	rw	R0 ... R7	R0 ... R7
GPR ₈₋₁₅	rw	R8 ... R15	R8 ... R15
RFLAGS	rw	RFLAGS*	RFLAGS
RIP	rw	RIP	RIP, Instruction Length, Instruction Info
STA	rw	-	Interruptibility State, Activity State
QUAL	r	Qualifications [†]	Qualifications [‡]
CTRL	w	-	Execution Controls, CR Intercepts EXC Intercepts, PF Error Mask/Match
TPR	w	-	TPR Threshold
INJ	rw r	-	Interruption Info, Interruption Error IDT Vectoring Info, IDT Vectoring Error
CS/SS	rw	-	CS, SS (Selector, Base, Limit, AR)
DS/ES	rw	-	DS, ES (Selector, Base, Limit, AR)
FS/GS	rw	-	FS, GS (Selector, Base, Limit, AR)
TR	rw	-	TR (Selector, Base, Limit, AR)
LDTR	rw	-	LDTR (Selector, Base, Limit, AR)
GDTR	rw	-	GDTR (Base, Limit)
IDTR	rw	-	IDTR (Base, Limit)
PDPTE	rw	-	PDPTE{0-3}
CR	rw	-	CR0, CR2, CR3, CR4, CR8
DR	rw	-	DR7
XSAVE	rw	-	XCR0, IA32_XSS
SGX	rw	-	IA32_SGXLEPUBKEYHASH{0-3}
APIC	w	-	IA32_APIC_BASE [§]
SYSCALL	rw	-	IA32_{STAR, LSTAR, FMASK}
SYSENTER	rw	-	IA32_SYSENTER_{CS, ESP, EIP}
PAT	rw	-	IA32_PAT
EFER	rw	-	IA32_EFER
KERNEL_GS	rw	-	IA32_KERNEL_GS_BASE
TSC	rw	-	IA32_TSC_AUX
TLB	w	-	Invalidates the TLB for the vCPU
SPACES	w	-	SEL_GST, SEL_PIO, SEL_MSR

*Only the status and control flags are writable.

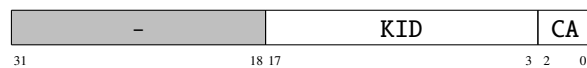
[†]Qualification fields contain exception error code (1st), page-fault linear address (2nd).

[‡]Qualification fields contain exit qualification (1st), guest-linear address (2nd), guest-physical address (3rd).

[§]APIC access page guest-physical address in the vCPU's currently assigned guest space.

8.6.5 Memory Attribute Descriptor

The Memory Attribute Descriptor (MAD) describes page attributes for a CAP_{MEM} delegation from $\text{SPC}_{\text{HST}_{\text{NOVA}}}$.



The fields are defined as follows:

CA

Specifies the cacheability attributes of the page according to the following table:

Encoding	Cacheability	Description
0x0	WB	Write Back
0x1	WT	Write Through
0x2	WC	Write Combining
0x3	UC	Strong Uncacheable
0x4	WP	Write Protected
0x5	–	<i>reserved</i>
0x6	–	<i>reserved</i>
0x7	–	<i>reserved</i>

KID

Specifies the key identifier of the cryptographic key to be used for memory encryption – must be $\leq \text{KID}_{\text{MAX}}$.

Please refer to [4, 5] for the architectural behavior of these attributes.

8.6.6 Interrupt Semaphore Descriptor

The Interrupt Semaphore Descriptor (ISD) describes a GSI associated with an Interrupt Semaphore.



The fields are defined as follows:

S

PCI Segment

GSI

Designates the interrupt via its [GSI](#) number in the following range:

- $[0, \text{GSI}_{\text{NUM}})$ – Global System Interrupt (GSI)

8.6.7 Device Topology Descriptor

The Device Topology Descriptor (DTD) describes the topology of a device as follows:

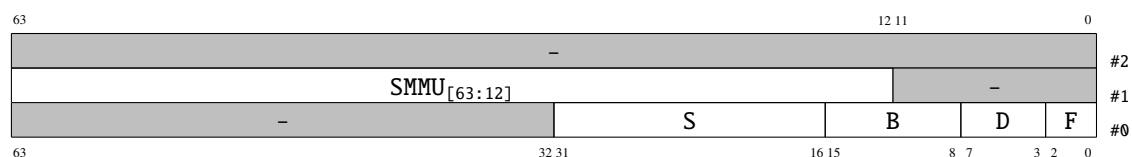
S, B, D, F

PCI Segment/Bus/Device/Function (SBDF).

SMMU

Host-Physical Address (HPA) of the SMMU that performs DMA and MSI translation for the device.

Reserved fields and fields that do not apply to a particular platform must be zero.



8.7 Calling Convention

The following pages describes the calling convention for each hypercall. A [Host Execution Context](#) calls into the microhypervisor by loading the hypercall identifier and other parameters into the specified [CPU](#) registers and then executes the `syscall` instruction [4, 5].

The hypercall identifier consists of the [hypercall number](#) and hypercall-specific flags, as illustrated in Figure 8.1.

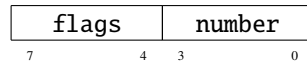


Figure 8.1: Hypercall Identifier

The [status code](#) returned from a hypercall has the format shown in Figure 8.2.

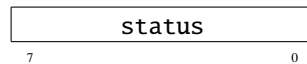
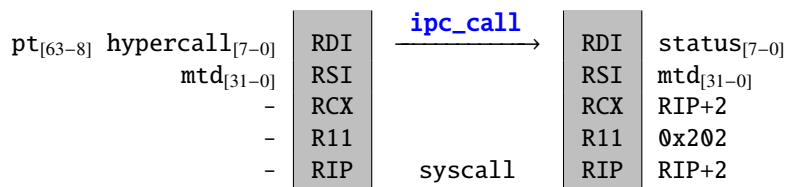


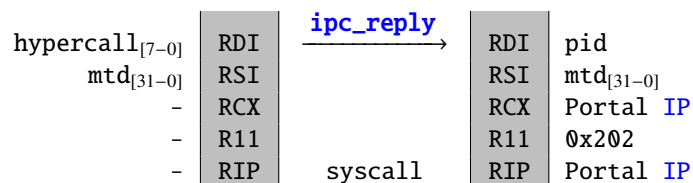
Figure 8.2: Status Code

The assignment of hypercall parameters to [CPU](#) registers is shown on the left side, the contents of the [CPU](#) registers after the hypercall is shown on the right side.

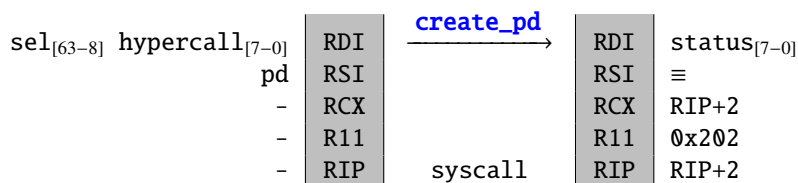
IPC Call



IPC Reply



Create Protection Domain



Create Execution Context

sel _[63-8]	hypercall _[7-0]	RDI	create_ec	RDI	status _[7-0]
	pd	RSI		RSI	≡
	hvp _[63-12]	RDX		RDX	≡
evt _[63-16]	cpu _[15-0]	RAX		RAX	≡
	sp	R8		R8	≡
	-	RCX		RCX	RIP+2
	-	R11		R11	0x202
	-	RIP	syscall	RIP	RIP+2

Create Scheduling Context

sel _[63-8]	hypercall _[7-0]	RDI	create_sc	RDI	status _[7-0]
	pd	RSI		RSI	≡
	ec	RDX		RDX	≡
	scd	RAX		RAX	≡
	-	RCX		RCX	RIP+2
	-	R11		R11	0x202
	-	RIP	syscall	RIP	RIP+2

Create Portal

sel _[63-8]	hypercall _[7-0]	RDI	create_pt	RDI	status _[7-0]
	pd	RSI		RSI	≡
	ec	RDX		RDX	≡
	ip	RAX		RAX	≡
	-	RCX		RCX	RIP+2
	-	R11		R11	0x202
	-	RIP	syscall	RIP	RIP+2

Create Semaphore

sel _[63-8]	hypercall _[7-0]	RDI	create_sm	RDI	status _[7-0]
	pd	RSI		RSI	≡
	val	RDX		RDX	≡
	-	RCX		RCX	RIP+2
	-	R11		R11	0x202
	-	RIP	syscall	RIP	RIP+2

Create Device Context

sel _[63-8]	hypercall _[7-0]	RDI	create_dc	RDI	status _[7-0]
	pd	RSI		RSI	≡
	dtd#0	RDX		RDX	≡
	dtd#1	RAX		RAX	≡
	dtd#2	R8		R8	≡
	-	RCX		RCX	RIP+2
	-	R11		R11	0x202
	-	RIP	syscall	RIP	RIP+2

Control Protection Domain

src _[63-8]	hypercall _[7-0]	RDI	→ ctrl_pd	RDI	status _[7-0]
	dst	RSI		RSI	≡
ssb _[63-12]	ord _[4-0]	RDX		RDX	≡
dsb _[63-12]	pmu _[5-0]	RAX		RAX	≡
	mad _[31-0]	R8		R8	≡
		RCX		RCX	RIP+2
		R11		R11	0x202
		RIP	syscall	RIP	RIP+2

Control Execution Context

ec _[63-8]	hypercall _[7-0]	RDI	→ ctrl_ec	RDI	status _[7-0]
		RCX		RCX	RIP+2
		R11		R11	0x202
		RIP	syscall	RIP	RIP+2

Control Scheduling Context

sc _[63-8]	hypercall _[7-0]	RDI	→ ctrl_sc	RDI	status _[7-0]
		RSI		RSI	stc
		RCX		RCX	RIP+2
		R11		R11	0x202
		RIP	syscall	RIP	RIP+2

Control Portal

pt _[63-8]	hypercall _[7-0]	RDI	→ ctrl_pt	RDI	status _[7-0]
	pid	RSI		RSI	≡
	mtid _[31-0]	RDX		RDX	≡
		RCX		RCX	RIP+2
		R11		R11	0x202
		RIP	syscall	RIP	RIP+2

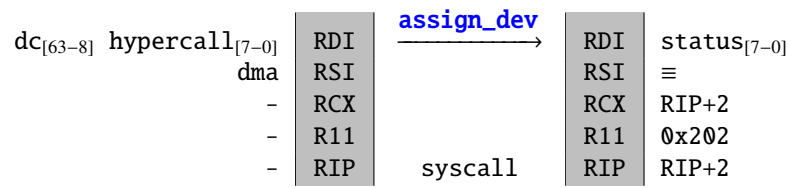
Control Semaphore

sm _[63-8]	hypercall _[7-0]	RDI	→ ctrl_sm	RDI	status _[7-0]
	stc	RSI		RSI	≡
		RCX		RCX	RIP+2
		R11		R11	0x202
		RIP	syscall	RIP	RIP+2

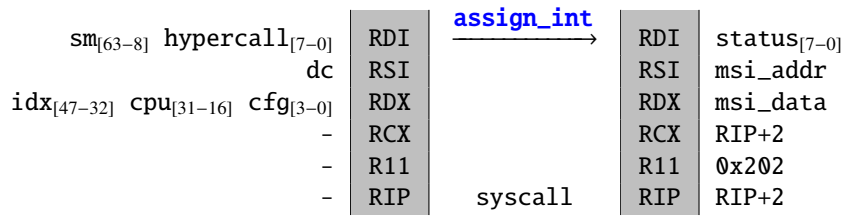
Control Hardware

desc _[63-8]	hypercall _[7-0]	RDI	→ ctrl_hw	RDI	status _[7-0]
		RCX		RCX	RIP+2
		R11		R11	0x202
		RIP	syscall	RIP	RIP+2

Assign Device



Assign Interrupt



Part V

Appendix

A Acronyms

ACPI	Advanced Configuration and Power Interface [7]
BSP	Bootstrap Processor
CAP	Capability
CAP₀	Null Capability
CAP_{OBJ}	Object Capability
CAP_{OBJOBJ}	Object Space Capability
CAP_{OBJHST}	Host Space Capability
CAP_{OBJGST}	Guest Space Capability
CAP_{OBJDMA}	DMA Space Capability
CAP_{OBJPIO}	PIO Space Capability
CAP_{OBJMSR}	MSR Space Capability
CAP_{OBJPD}	PD Capability
CAP_{OBJEC}	EC Capability
CAP_{OBJSC}	SC Capability
CAP_{OBJPT}	PT Capability
CAP_{OBJSM}	SM Capability
CAP_{OBJDC}	DC Capability
CAP_{MEM}	Memory Capability
CAP_{PIO}	PIO Capability
CAP_{MSR}	MSR Capability
CAT	Cache Allocation Technology [4]
CDP	Code and Data Prioritization [4]
COS	Class Of Service [Arm, x86] [4]
CPU	Central Processing Unit [3, 4, 5]
CRTM	Core Root of Trust for Measurement [21, 22]
CTX	Translation Context [12, 13]
DID	Device Identifier [11]
DMA	Direct Memory Access
DPR	DMA Protected Range
DRTM	Dynamic Root of Trust for Measurement [21, 22]
DTD	Device Topology Descriptor [Arm, x86]
DC	Device Context
DVA	DMA-Virtual Address
EC	Execution Context
EC_{HST}	Host Execution Context
EC_{GST}	Guest Execution Context
EC_{CURRENT}	Current Execution Context

EC_{ROOT}	Root Execution Context
ELF	Executable and Linkable Format [23]
ESPI	Extended Shared Peripheral Interrupt [10, 11]
FDT	Flattened Device Tree [24]
FPU	Floating Point Unit [3, 4, 5]
GIC	Generic Interrupt Controller [10, 11]
GICC	GIC CPU Interface
GICD	GIC Distributor
GICH	GIC HYP Interface
GICR	GIC Redistributor
GITS	GIC Interrupt Translation Service
GPA	Guest-Physical Address
GSI	Global System Interrupt [7]
HIP	Hypervisor Information Page [Arm, x86]
HPA	Host-Physical Address
HVA	Host-Virtual Address
IOAPIC	I/O Advanced Programmable Interrupt Controller
IOMMU	I/O Memory Management Unit [17, 18]
IP	Instruction Pointer
IPC	Inter-Process Communication
ISD	Interrupt Semaphore Descriptor [Arm, x86]
KID	Key Identifier
LAPIC	Local Advanced Programmable Interrupt Controller
LIM	Launch Integrity Measurement
LPI	Locality-Specific Peripheral Interrupt [10, 11]
MAD	Memory Attribute Descriptor [Arm, x86]
MBA	Memory Bandwidth Allocation [4]
MMU	Memory Management Unit [3, 4, 5]
MSI	Message-Signaled Interrupt [25, 26]
MSR	Model-Specific Register [4, 5]
MTD	Message Transfer Descriptor [Arm, x86]
NOVA	NOVA OS Virtualization Architecture [2]
PCI	Peripheral Component Interconnect [25, 26]
PCR	Platform Configuration Register [21, 22]
PIO	Port-Based I/O
PD	Protection Domain
PD_{ROOT}	Root Protection Domain
PID	Portal Identifier
PT	Portal
QOS	Quality Of Service
RIM	Reference Integrity Measurement
SBDF	PCI Segment/Bus/Device/Function

SC	Scheduling Context
SC_{CURRENT}	Current Scheduling Context
SC_{ROOT}	Root Scheduling Context
SCD	Scheduling Context Descriptor
SEL	Capability Selector
SEL_{EVT}	Event Selector Base [Arm , x86]
SEL_{OBJ}	Object Capability Selector
SEL_{HST}	Host Capability Selector
SEL_{GST}	Guest Capability Selector
SEL_{DMA}	DMA Capability Selector
SEL_{PIO}	PIO Capability Selector
SEL_{MSR}	MSR Capability Selector
SID	Stream Identifier [12 , 13]
SM	Semaphore
SMG	Stream Mapping Group [12 , 13]
SMMU	System Memory Management Unit [12 , 13]
SP	Stack Pointer
SPC_{OBJ}	Object Space
SPC_{HST}	Host Space
SPC_{GST}	Guest Space
SPC_{DMA}	DMA Space
SPC_{PIO}	PIO Space
SPC_{MSR}	MSR Space
SPC_{OBJCURRENT}	Current Object Space
SPC_{OBJNOVA}	NOVA Object Space
SPC_{HSTNOVA}	NOVA Host Space
SPC_{PIONOVA}	NOVA PIO Space
SPC_{MSRNOVA}	NOVA MSR Space
SPC_{OBJROOT}	Root Object Space
SPC_{HSTROOT}	Root Host Space
SPC_{PIOROOT}	Root PIO Space
SPI	Shared Peripheral Interrupt [10 , 11]
STC	System Time Counter
TPM	Trusted Platform Module [21 , 22]
TXT	Trusted Execution Technology [19]
UART	Universal Asynchronous Receiver Transmitter
UEFI	Unified Extensible Firmware Interface [8]
UTCB	User Thread Control Block [Arm , x86]
VMM	Virtual-Machine Monitor

ipc_call	Hypercall [Arm, x86]: IPC Call
ipc_reply	Hypercall [Arm, x86]: IPC Reply
create_pd	Hypercall [Arm, x86]: Create Protection Domain
create_ec	Hypercall [Arm, x86]: Create Execution Context
create_sc	Hypercall [Arm, x86]: Create Scheduling Context
create_pt	Hypercall [Arm, x86]: Create Portal
create_sm	Hypercall [Arm, x86]: Create Semaphore
create_dc	Hypercall [Arm, x86]: Create Device Context
ctrl_pd	Hypercall [Arm, x86]: Control Protection Domain
ctrl_ec	Hypercall [Arm, x86]: Control Execution Context
ctrl_sc	Hypercall [Arm, x86]: Control Scheduling Context
ctrl_pt	Hypercall [Arm, x86]: Control Portal
ctrl_sm	Hypercall [Arm, x86]: Control Semaphore
ctrl_hw	Hypercall [Arm, x86]: Control Hardware
assign_dev	Hypercall [Arm, x86]: Assign Device
assign_int	Hypercall [Arm, x86]: Assign Interrupt

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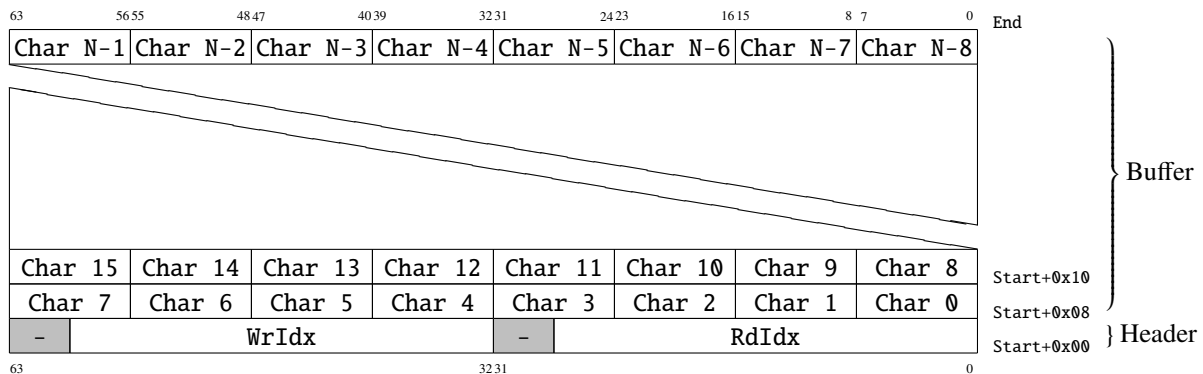
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C Console

C.1 Memory-Buffer Console

The [NOVA](#) microhypervisor implements a memory-buffer console that provides run-time debug output. The memory-buffer console consists of a signaling semaphore (see [6.1.3.1](#)) and an in-memory data structure with a header and a buffer as follows:



The start address and end address of the memory-buffer console are conveyed in the [HIP](#).

The buffer size (N characters) can be computed as:

$$N = \text{MBUF End Address} - \text{MBUF Start Address} - \text{MBUF Header Size}$$

The fields of the header are used as follows:

- **RdIdx** is in range $[0, N-1]$.
It points to the **next** character in the buffer that the console consumer will read and is typically advanced by the console consumer.
- **WrIdx** is in range $[0, N-1]$.
It points to the **next** character in the buffer that the [NOVA](#) microhypervisor will write and is only advanced by the [NOVA](#) microhypervisor.
- The buffer is empty if **RdIdx** is equal to **WrIdx**.
- Otherwise **WrIdx** is ahead of **RdIdx**, wrapping around the buffer size N accordingly, i.e. character $N+x$ will be stored in the same buffer slot as character x .
- If the buffer becomes full, the [NOVA](#) microhypervisor advances **RdIdx**, forcing the oldest character to be discarded from the buffer.
- At the end of each line, the [NOVA](#) microhypervisor invokes [ctrl_sm](#) (Up) on the signaling semaphore. The console consumer should use [ctrl_sm](#) (Down) on the signaling semaphore instead of polling **WrIdx**.

C.2 UART Console

Additionally several different [UART](#) consoles can be used to provide boot-time-only debug output of the microhypervisor. [UART](#) consoles must be configured for 115200 baud and 8N1 mode.

D Download

The source code of the [NOVA](#) microhypervisor and the latest version of this document can be downloaded from GitHub: <https://github.com/udosteinberg/NOVA>